

From Geometric Correction Orders to Detector-Centered Phase

Exact Middle Transfer Bridge, Compactness Response, and Unified-Dynamics
Constraints

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Manuscript status. Technical research paper / derivation note. Not peer reviewed. Exact statements are exact only under their explicit hypotheses; source-locked model statements are not promoted to fundamental derivations; regression and technical checks are not counted as independent empirical confirmations. For vacuum electromagnetism the paper uses standard covariant Maxwell dynamics on the declared static SSZ background, so the eikonal/null-Hamiltonian/Fermat route is derived on that background. The canonical weak-field observable layer remains source-locked to $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$; deriving that completion from a single unified action remains open. The paper does not claim a unique fundamental SSZ action, a globally regular interior, rotation/dynamical response, electromagnetic backreaction, plasma/material constitutive closure, or independent empirical confirmation of a new strong-field discriminator.

Validation summary. Before the compactness-response extension was integrated, the available strong-field audit was re-executed successfully and reproduced its typed ledger: 18 PASS, 4 structural FLAG, 5 REJECTED_CANDIDATE, 1 CONDITIONAL_PASS, 1 OPEN_CANDIDATE, 1 REGULARITY_FLAG, and 1 INTERPRETIVE_RESULT. The lower Middle-Bridge proof/test lock remains 35/35 PASS and the independent consistency wrapper remains 15/15. The new compactness-response audit adds 13/13 PASS checks covering the exact variable transformation, pure-branch flow identities, transformed C^2 matching, the mean-value susceptibility bound, global polynomial extrema, and an independent numerical quadrature/optimization cross-check. These layers remain typed and are not added into a single empirical score.

Abstract

This paper joins the full exact Middle Transfer Bridge with the strongest corrected strong-field dynamics results and a new compactness-response audit. At the scalar phase level, an absolutely convergent lifted phase decomposition $\Phi = \sum_n \Delta\Phi_n$ maps exactly to multiplicative $U(1)$ transfer, $\prod_n e^{i\Delta\Phi_n} = e^{i\Phi}$, and the continuous scalar generator satisfies $U' = iaU$, hence $U = \exp[i \int a ds]$. For the declared Sagnac-type local law $dt_\sigma = ds/[c(1 - \sigma\beta)]$, the Neumann hierarchy is an exact conditional corollary for $|\beta| < 1$; spatial path segmentation j and correction order n remain independent indices, and counter-propagating parity removes even orders from the antisymmetric difference. The paper keeps the finite- β frame/clock split explicit and distinguishes correction-order convergence from spatial quadrature convergence.

For vacuum electromagnetic propagation on the declared static SSZ metric, standard covariant Maxwell theory yields the WKB eikonal equation $g^{\mu\nu}k_\mu k_\nu = 0$, the null Hamiltonian, the optical/Fermat formulation, and a detector endpoint-delay phase $U_\gamma = e^{i\omega\Delta T_\gamma}$ under the stated Fourier convention. A local static observer measures c ; coordinate travel time and local light speed are not conflated. The full spacetime eikonal is constant along its own null ray, so the detector transfer phase is not a photon proper-time accumulation. Killing frequency, local detector frequency, source/node phases, and detector reference phase are kept as typed quantities.

The unified-dynamics audit uses the current public/static P0 profile: inner g_2 , $\Xi_s = 1 - e^{-\varphi/x}$ for $1 \leq x < 1.8$; a derivative-matched quintic C^2 transition on $1.8 \leq x \leq 2.2$; and outer g_1 , $\Xi_o = 1/(2x)$ for $x > 2.2$. Formula domain is separated from physical regime: the project taxonomy still labels $2.2 < x \leq 3$ photon-sphere context, $3 < x \leq 10$ physical strong field, and only $x > 10$

physical weak field. The inner branch carries the differential fingerprint $x^2\Xi'_s + \varphi(1 - \Xi_s) = 0$; the outer branch obeys $x\Xi'_o + \Xi_o = 0$ and $2x\Xi_o = 1$. Reversal of the fixed-endpoint Hermite bridge is mathematically neutral; “strong-first” is therefore a research orientation, not a direction built into the interpolation theorem.

Writing compactness as $C = r_s/r = 1/x$ and $u = GM/(rc^2) = C/2$ gives an exact reparameterization of the pure branches: g_1 becomes $\Xi_o = u$, while g_2 becomes $\Xi_s = 1 - e^{-2\varphi u}$. Their pure-branch flow laws are $u d\Xi_o/du = \Xi_o$ and $d\Xi_s/du = 2\varphi(1 - \Xi_s)$. The fixed transition window maps to $u \in [5/22, 5/18]$. Any differentiable bridge with the canonical endpoint values must, by the mean-value theorem, attain $d\Xi/du = 7.241123407889$ somewhere, already far above the pure-branch endpoint susceptibilities 1 and 1.317124872442. The canonical Hermite bridge has a globally certified numerical maximum $\max d\Xi/du = 12.688510828116$ at $x = 2.008739217432$. This susceptibility amplification is a property of the fixed endpoint values and narrow transition window; interpreting it as a physical response mechanism remains an open dynamical hypothesis.

Reconstructing the Einstein-effective source gives $\hat{p}_r = -\hat{\rho}$, $\hat{\rho} = 2\hat{m}'/x^2$, and $\hat{p}_t = -\hat{m}''/x$. The canonical endpoint masses, $\hat{m}(1.8) = 0.545334802123639$ and $\hat{m}(2.2) = 0.369684499314129$, imply the interpolation-independent identity $\int_{1.8}^{2.2} x^2 \hat{\rho} dx = -0.351300605619020$ and force $\hat{\rho} \leq -0.218833433733$ somewhere for every differentiable Einstein-matter bridge with the same endpoint metric values. The strong branch reaches $\hat{m} = 1/2$ already at $x_* = 1.594811457889$, while the outer branch tends to $1/2$ from below. With the outer endpoint fixed at $x_1 = 2.2$, positive density requires $x_0 \leq 1.082828978184$; the stronger tangential-NEC monotonicity gate requires $x_0 \leq 0.977946246457$, outside the frozen exterior. Thus the current endpoint architecture cannot be sourced by an everywhere-NEC ordinary Einstein-matter transition inside $x \geq 1$.

Constrained mass-space constructions show that nonnegative density can be recovered after moving the handoff to $x = 1$, with global critical-point numerical certificates for the stored floating-point polynomials, but the fixed- x_1 NEC theorem prevents an ordinary everywhere-NEC rescue within the frozen exterior. Candidate-source filters reject, as sole exact sources, minimally coupled static radial scalar sums, perfect fluid, pure linear Maxwell, linear Maxwell plus constant Λ , and the simplest uncoupled global-monopole plus fixed-charge Maxwell model. A corrected magnetic-NED diagnostic retains in-domain characteristic-sign zero crossings in the canonical transition, while the pure branches admit a positive defect-like plus gauge-like algebraic decomposition. This motivates generalized hedgehog/gauge, NED-coupled, or second-order/degenerate scalar-vector-tensor dynamics, but no unique action is claimed. The canonical observable PPN layer remains source-locked to $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$, while its derivation from the same unified dynamics remains open. The decisive next step is a concrete covariant action that derives the strong branch, generates its own transition, recovers the canonical PPN limit, passes quadratic stability, and yields a sealed prospective strong-field discriminator before target data are opened.

Keywords: Sagnac effect; phase transfer; $U(1)$; Neumann series; interferometry; detector-centered phase; JIF; Segmented Spacetime; forward prediction; anti-circular validation.

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1 Scope, source lock, and claim boundary

This paper is source-locked to the current project architecture and to a public Forward-Lab snapshot rather than to remembered intermediate drafts. The relevant sources are: the integrated JIF manuscript; the current SSZ regime documentation; the variational C^2 bridge manuscript; the FFT phase-segmentation manuscript; the May 2026 geometric Sagnac correction-series preprint; and the archived public repository `error-wtf/ssz-jif-forward-lab`. For the electromagnetic coupling layer, the final edition retains the source lock to the supplied SSZ helper corpus: the expanded integrated edition, the “Radial Scaling Gauge for Maxwell Fields” note, and the “Maxwell Waves as Rotating Space” note. Those sources state that practical electromagnetic calculations retain Maxwell theory, that the geometric-optics limit follows null geodesics of the effective metric, and that the radial scaling picture is a geometric reparameterization with invariant locally measured c [18, 19, 20]. The repository audit in this revision is tied to commit `90eb4eb594ca41f8ca92d3c4945b108d23a0dc5c` (5 August 2026), whose preceding curated research snapshot is `6596d64db36bef7c9a7e1fe2529872cbfd2e8a8a`. For weak-field observable routing, the final edition additionally locks to the public `ssz-complete-documentation` files `03_FORMULAS/ppn_formulas.md`, `11_GUARDRAILS/prime_directive.md`, and `11_GUARDRAILS/method_assignment.md` at the referenced repository snapshot `b2bac976a75dd70f26599b7debd19d88a835f429`, and to the `implementation/tests` in `SSZ-METRIC-COMPLETE` at snapshot `b7643f437f46a170f05e492609b27ed2a8acae73`. Those sources explicitly lock $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ and route light-path observables through PPN completion rather than a Ξ -only shortcut [21, 22, 23, 24].

Canonical P0 and formula-domain lock. For the static exterior audit used in this paper,

$$\Xi_s(x) = 1 - e^{-\varphi/x}, \quad 1 \leq x < 1.8, \quad (1)$$

$$\Xi_o(x) = \frac{1}{2x}, \quad x > 2.2, \quad (2)$$

with the derivative-matched quintic C^2 Hermite bridge on $1.8 \leq x \leq 2.2$, $D = (1 + \Xi)^{-1}$, $A = D^2$, $x = r/r_s$, $r_s = 2GM/c^2$, and $\varphi = (1 + \sqrt{5})/2$. The theory-readiness freeze treats $x \geq 1$ as the exterior comparison domain. The project regime taxonomy separately labels $2.2 < x \leq 3$ photon-sphere context, $3 < x \leq 10$ physical strong field, and $x > 10$ physical weak field. Calculations below $x = 1$ are therefore formal continuation diagnostics, not frozen-exterior claims. The canonical observable PPN layer remains $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ as the source-locked weak-field completion; deriving it from the same unified dynamics remains an open structural task.

The paper uses three evidence labels informally:

- **Exact:** follows as mathematics from explicitly stated assumptions.
- **Regression/technical:** checks algebra, code semantics, numerical composition, or reference data processing.
- **Empirical forward evidence:** requires a frozen physical prediction compared with an independent target under the declared anti-circular protocol.

Only the first two categories are claimed here.

Upstream validation context. Within the derivations below, the SSZ geometry is an input: this paper does not re-derive the background from a fundamental action. That does not mean the upstream geometry/clock stack is untested. The frozen cross-repository snapshot in `ssz-all-tests`

separately exercises the $\Xi \rightarrow D, s$, metric, geodesic/PPN, clock/frequency and phase-adjacent layers across heterogeneous regression, benchmark and observational-data tests. The present paper maps those upstream checks onto the architecture in section 20.2. They support software/model consistency of the inputs but are not used as premises in the proofs of the $U(1)$ factorization or scalar-generator theorems.

2 Notation and reference discipline

A recurring source of confusion is that “time”, “path”, and “phase” each have multiple physically distinct representatives. We therefore fix the notation before deriving anything.

Table 1: Core variables and their roles.

Symbol	Type	Meaning
s	path coordinate	declared optical/null/path parameter used by the local transfer law
t_{ax}	coordinate time	time of the non-rotating axle/laboratory frame in the ideal circular Sagnac construction
τ_D	proper time	clock time of the co-rotating detector
$x = r/r_s$	SSZ gravitational coordinate	black-hole-centric radial coordinate used by the canonical SSZ radial profile
z/z_0	jet coordinate	source-model propagation coordinate along a jet
R/R_0	plasmoid coordinate	source-size/expansion coordinate of an expanding plasmoid
ω_t	coordinate-frequency	phase rate conjugate to a chosen coordinate time t
ω_D	detector frequency	phase rate conjugate to detector proper time τ_D
Φ	lifted phase	real, unwrapped phase when cycle history is retained
$e^{i\Phi}$	$U(1)$ phase	wrapped scalar phase factor, insensitive to integer multiples of 2π

Conjugate-clock rule. A product ωdt is meaningful as a phase increment only when the angular frequency ω is conjugate to the same time coordinate as dt . If the clock variable is changed, the frequency must transform so that the scalar phase increment remains the same:

$$d\Phi = \omega_t dt = \omega_D d\tau_D.$$

This rule is the clean way to prevent a hidden extra or missing Lorentz factor.

3 Operational architecture: where the middle bridge sits

The JIF program starts from counted phase,

$$dJ = \frac{d\Phi}{2\pi}.$$

For a massive or bound oscillator with proper frequency f_0 , ordinary phase evolution gives

$$d\Phi = 2\pi f_0 d\tau, \quad dJ = f_0 d\tau.$$

Within the canonical static SSZ chain, $\Xi(r)$ defines $D(r) = 1/(1 + \Xi(r))$, and a selected static observer family uses $d\tau = D(r)dT$. This timelike branch is already operationally closed once the observer family and oscillator are specified.

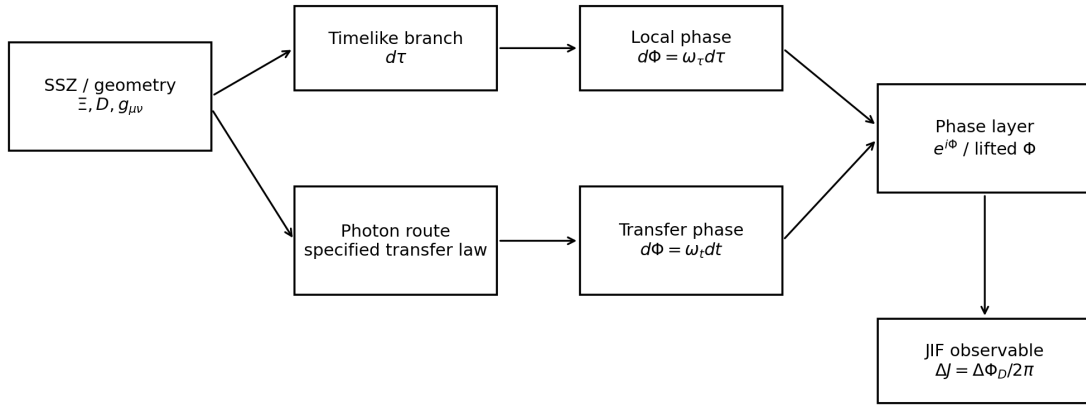
The photon branch is different. A null signal is not assigned a timelike rest-frame clock. The current detector-centered JIF construction instead assembles the complete phase contribution at a common detection event,

$$\Phi_D^{(a)} = \Phi_E^{(a)} + \Phi_{\text{transfer}}^{(a)} + \sum_m \Phi_{\text{node},m}^{(a)} + \Phi_{\text{detector}}^{(a)}, \quad (3)$$

and only then forms

$$\Delta J_D = \frac{\Phi_D^{(b)} - \Phi_D^{(a)}}{2\pi}. \quad (4)$$

The architectural question solved in this paper is therefore: given a specified local transfer law, how does it become the transfer phase in equation (3)?



Different physical middle layers meet only at the detector-referenced phase layer.

Figure 1: Operational middle bridge. The timelike and photon branches use different physical middle layers and meet only at a referenced phase layer.

4 A general exact $U(1)$ bridge theorem

The most robust form of the middle bridge does not depend on Sagnac geometry.

Theorem 1 (additive phase to multiplicative $U(1)$ transfer). Let $\{\Delta\Phi_n\}_{n \geq 0} \subset \mathbb{R}$ satisfy

$$\sum_{n=0}^{\infty} |\Delta\Phi_n| < \infty.$$

Define $T_n = \exp(i\Delta\Phi_n)$. Then the infinite product converges and

$$\prod_{n=0}^{\infty} T_n = \exp\left(i \sum_{n=0}^{\infty} \Delta\Phi_n\right). \quad (5)$$

Proof. The partial products satisfy identically

$$\prod_{n=0}^M e^{i\Delta\Phi_n} = e^{i \sum_{n=0}^M \Delta\Phi_n}.$$

Absolute convergence of the real phase series gives convergence of its partial sums to a finite limit;

continuity of the exponential gives equation (5). □

This is the group-homomorphism statement

$$\rho : (\mathbb{R}, +) \longrightarrow U(1), \quad \rho(\Phi) = e^{i\Phi}, \quad \rho(\Phi_1 + \Phi_2) = \rho(\Phi_1)\rho(\Phi_2).$$

The theorem does *not* claim that every summand is a physically separate propagation episode. A correction hierarchy can be a mathematical decomposition of one physical transfer. The physically observable object is the completed referenced phase.

4.1 Why the exponential appears: growth, rotation, and the generator

The notation $e^{i\Phi}$ is often introduced as a convenient shorthand for $\cos \Phi + i \sin \Phi$. For the present bridge it is more useful to state the dynamical reason. The real exponential is the unique normalized solution of

$$\frac{dy}{dx} = y, \quad y(0) = 1,$$

so $y = e^x$. Replacing the real generator 1 by the imaginary generator i gives

$$\frac{dU}{d\Phi} = iU, \quad U(0) = 1,$$

and therefore

$$\boxed{U(\Phi) = e^{i\Phi} = \cos \Phi + i \sin \Phi.}$$

Now the derivative is a quarter-turn of the state itself. In the real two-dimensional representation, multiplication by i maps (x, y) to $(-y, x)$, which is tangent to the circle through (x, y) . Consequently

$$\frac{d}{d\Phi} |U|^2 = 2 \operatorname{Re} \left(U^* \frac{dU}{d\Phi} \right) = 2 \operatorname{Re}(i|U|^2) = 0,$$

so $|U| = 1$ for all Φ . The exponential therefore changes role: e^x describes self-proportional growth along the real direction, while $e^{i\Phi}$ describes norm-preserving rotation along the imaginary direction.

In Lie-group language, $U(1) = \{z \in \mathbb{C} : |z| = 1\}$ has Lie algebra $\mathfrak{u}(1) = i\mathbb{R}$, and the exponential map sends the additive generator $i\Phi$ to the group element $e^{i\Phi}$. This is the structural reason that additive phase increments become multiplicative transfer factors.

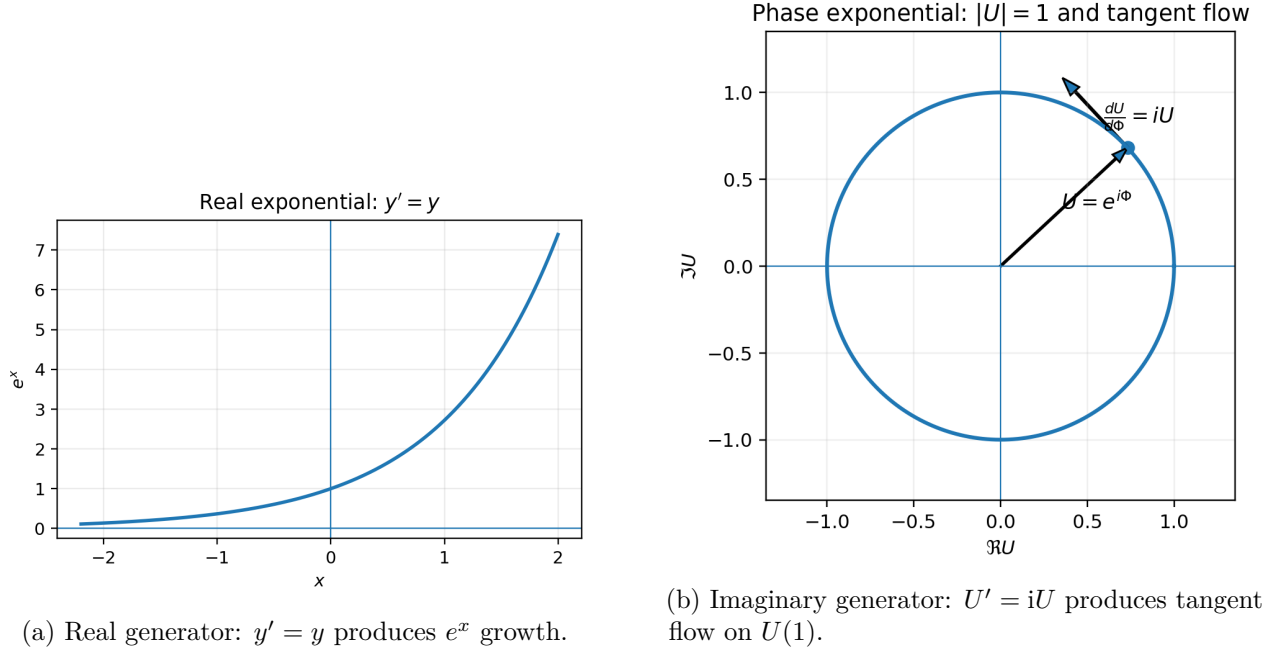


Figure 2: The same exponential function describes growth or phase rotation depending on the generator. The Middle Bridge uses the unitary, imaginary-generator case.

Three standard appearances of e should therefore be kept distinct:

Table 2: Three mathematically related but logically distinct appearances of the exponential.

Construction	Formula	Role in this paper
Compound-growth limit	$\lim_{N \rightarrow \infty} (1 + x/N)^N = e^x$	useful intuition, but not the proof of the bridge
Power-series definition	$e^z = \sum_{m \geq 0} z^m / m!$	analytic definition of the exponential itself
Phase representation / flow	$U = e^{i\Phi}, dU/d\Phi = iU$	the actual $U(1)$ mechanism used by the bridge

4.2 Continuous generator formulation and partition independence

The homomorphism theorem has an equivalent differential form that makes the passage from local transfer to a continuum propagator explicit.

Theorem 2 (scalar phase-flow propagator). Let $a \in L^1([s_0, s_1], \mathbb{R})$ and let U satisfy

$$\frac{dU}{ds} = ia(s)U(s), \quad U(s_0) = 1.$$

Then the absolutely continuous solution is unique and is

$$U(s_1, s_0) = \exp \left[i \int_{s_0}^{s_1} a(s) ds \right]. \quad (6)$$

It obeys the exact composition law

$$U(s_2, s_0) = U(s_2, s_1)U(s_1, s_0),$$

and $|U| = 1$. For any partition whose Riemann sums converge to the integral,

$$\prod_j e^{ia(s_j^*)\Delta s_j} = e^{i\sum_j a(s_j^*)\Delta s_j} \longrightarrow \exp\left[i \int a(s)ds\right].$$

Thus the discrete product and the continuous solution are two representations of the same scalar phase accumulation. No noncommutative path ordering is required because the generator is scalar.

Proof. Define $A(s) = \int_{s_0}^s a(\lambda)d\lambda$. Then $U = e^{iA}$ is absolutely continuous and satisfies $U' = iA'e^{iA} = iA'U$ almost everywhere. Uniqueness follows from the linear first-order initial-value problem. Composition follows from additivity of the integral. The product statement is the finite homomorphism identity followed by convergence of the Riemann sum. $\square$

4.3 Dimensional discipline of the exponent

The exponent of an exponential must be dimensionless. Therefore Φ is an angle in radians, $a(s) = d\Phi/ds$ has inverse-path units, and ωdt is valid only when ω is the angular frequency conjugate to the same time variable as dt . This simple dimensional statement is the local version of the conjugate-clock rule used later in the finite- β frame audit.

4.4 Lifted phase and JIF

The map $\mathbb{R} \rightarrow U(1)$ is many-to-one because $e^{i(\Phi+2\pi m)} = e^{i\Phi}$. A long-duration JIF record therefore uses a continuous lift $\tilde{\Phi} \in \mathbb{R}$ or an equivalent cycle counter:

$$J = \frac{\tilde{\Phi}}{2\pi}.$$

Cycle-slip control is not optional; a wrapped complex phase alone cannot recover the integer history.

5 Sagnac-type Neumann hierarchy as a corollary

Consider a deliberately restricted kinematic transfer law

$$dt_\sigma = \frac{ds}{c(1 - \sigma\beta)}, \quad \sigma \in \{+1, -1\}, \quad |\beta| < 1. \quad (7)$$

For a general routed problem, β should be read as the declared directional kinematic quantity entering this law; it is not automatically a global SSZ field variable.

Because

$$\frac{1}{1 - \sigma\beta} = \sum_{n=0}^{\infty} (\sigma\beta)^n,$$

we obtain

$$dt_{\sigma,n} = \frac{ds}{c}(\sigma\beta)^n, \quad dt_\sigma = \sum_{n=0}^{\infty} dt_{\sigma,n}. \quad (8)$$

If ω_t is conjugate to the chosen time variable t , define $k_t = \omega_t/c$. Then

$$d\Phi_{\sigma,n} = \omega_t dt_{\sigma,n} = k_t ds(\sigma\beta)^n, \quad (9)$$

and

$$T_{\sigma,n} = \exp[ik_t ds(\sigma\beta)^n]. \quad (10)$$

By Theorem 1,

$$\prod_{n=0}^{\infty} T_{\sigma,n} = \exp \left[i \frac{k_t ds}{1 - \sigma\beta} \right]. \quad (11)$$

The exponential is not generated by a Trotter limit here. The geometric series reconstructs an additive transfer quantity; the exponential appears because additive phase is represented multiplicatively by $U(1)$.

5.1 Absolute and uniform convergence

For $|\beta| < 1$,

$$\sum_{n=0}^{\infty} |d\Phi_{\sigma,n}| \leq |k_t ds| \sum_{n=0}^{\infty} |\beta|^n = \frac{|k_t ds|}{1 - |\beta|}.$$

If $|\beta(s)| \leq \beta_{\max} < 1$ along a finite path, the Weierstrass M-test gives uniform convergence, which justifies exchanging the correction-order sum with the path integral under the usual regularity assumptions.

6 Spatial composition and the two-index (j, n) lattice

Partition the physical route into spatial segments $j = 1, \dots, N$ with lengths Δs_j . Define

$$\Delta\Phi_{j,n} = k_t \Delta s_j (\sigma\beta_j)^n. \quad (12)$$

Then

$$\Phi_{\text{transfer}}^{(N,M)} = \sum_{j=1}^N \sum_{n=0}^M \Delta\Phi_{j,n},$$

and

$$U_{\text{transfer}}^{(N,M)} = \prod_{j=1}^N \prod_{n=0}^M e^{i\Delta\Phi_{j,n}}.$$

The j and n axes have different meanings: j changes physical location; n changes correction order at fixed location. Under the uniform convergence bound, the order of these scalar sums/products may be exchanged.

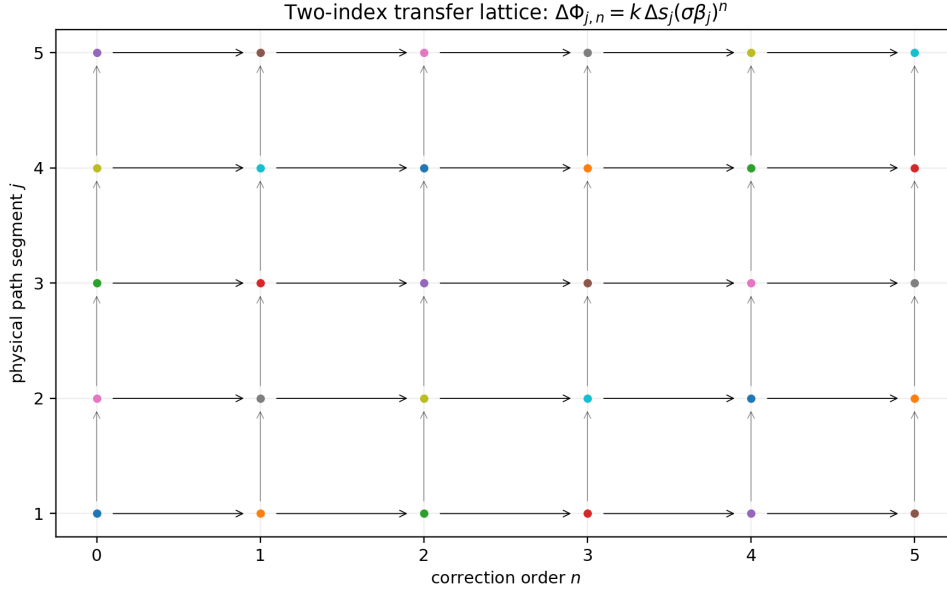


Figure 3: The two-index transfer lattice. Conflating j with n conflates physical discretization error with geometric-series truncation error.

For sufficiently regular $\beta(s)$,

$$U_\sigma = \exp\left[ik_t \int_\gamma \frac{ds}{1 - \sigma\beta(s)}\right]. \quad (13)$$

The scalar generator is $A_\sigma(s) = ik_t/[1 - \sigma\beta(s)]$. A path-ordering symbol is redundant because the scalar generators commute. Nontrivial path ordering appears only after introducing a genuinely matrix-valued internal space.

7 Counter-propagating pair and parity

For constant β and an axle-frame path length L ,

$$t_+ = \frac{L}{c(1 - \beta)}, \quad t_- = \frac{L}{c(1 + \beta)}.$$

Thus the axle-frame coordinate-time difference is

$$\Delta t_{\text{ax}} = \frac{2L\beta}{c(1 - \beta^2)}. \quad (14)$$

The Neumann difference is

$$\frac{1}{1 - \beta} - \frac{1}{1 + \beta} = 2(\beta + \beta^3 + \beta^5 + \cdots). \quad (15)$$

All even orders cancel. This is exactly the odd part under $\beta \mapsto -\beta$.

7.1 Circular specialization: the historical finite- β fix

For a circular path, $L = 2\pi R$, $v = \Omega R$, $A = \pi R^2$, and $\beta = \Omega R/c$. Then

$$\Delta t_{\text{ax}} = \frac{4A\Omega}{c^2}\gamma^2, \quad \gamma = (1 - \beta^2)^{-1/2}. \quad (16)$$

This is the exact return-time difference in the non-rotating axle frame for the idealized flat-spacetime moving-closure construction. It is not yet the proper-time reading of the rotating detector.

The detector clock satisfies

$$d\tau_D = dt_{\text{ax}} \sqrt{1 - \beta^2} = \frac{dt_{\text{ax}}}{\gamma},$$

so the exact detector proper-time difference is

$$\boxed{\Delta\tau_D = \frac{4A\Omega}{c^2} \gamma.} \quad (17)$$

This is the finite- β correction already distinguished in the May 2026 correction-series preprint and in standard relativistic Sagnac treatments. The leading term of both is

$$\Delta t_{\text{ax}} \simeq \Delta\tau_D \simeq \frac{4A\Omega}{c^2}, \quad \beta \ll 1.$$

7.2 Phase consistency across frames

If ω_D is the proper angular frequency of the detector-local reference oscillator, then the same phase advance written in axle coordinate time uses $\omega_{\text{ax}} = \omega_D/\gamma$:

$$\boxed{\omega_{\text{ax}} \Delta t_{\text{ax}} = \omega_D \Delta\tau_D.} \quad (18)$$

Accordingly, a detector-centered exact phase can be written as

$$\Delta\Phi_D = \omega_D \Delta\tau_D = \frac{4\omega_D A\Omega}{c^2} \gamma, \quad (19)$$

whereas $\omega_{\text{ax}} \Delta t_{\text{ax}}$ is the equivalent coordinate representation. The earlier draft's unqualified finite- β expression proportional to $(1 - \beta^2)^{-1}$ must therefore be read only as coordinate-time bookkeeping, not automatically as detector proper phase.

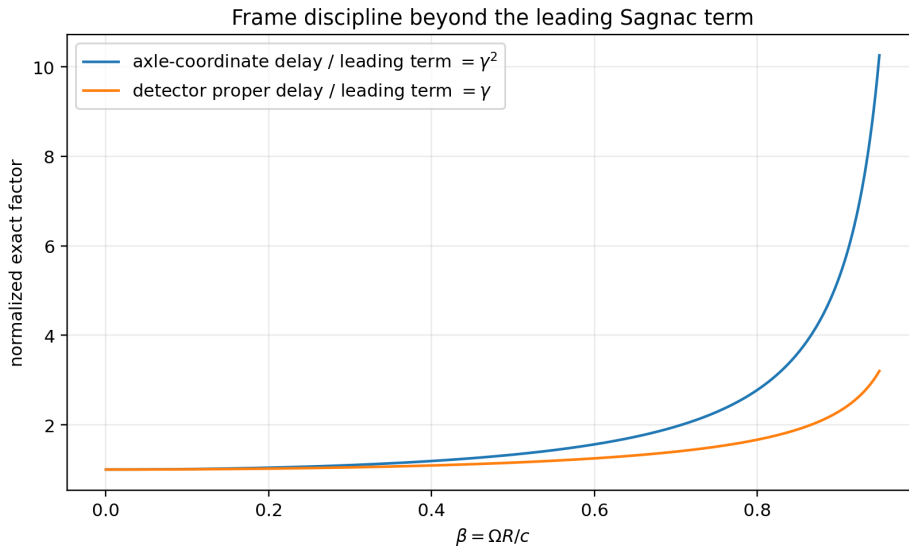


Figure 4: Beyond leading order, coordinate return time and detector proper time carry different relativistic factors. The phase remains invariant when the conjugate frequency is transformed consistently.

Scope of the exact circular formulas. These expressions assume an idealized prescribed circular path in flat spacetime. At high peripheral speed, rotating-ring rigidity, material response, refractive media, path definition, and detector protocol must be specified. The paper therefore uses the exact factors primarily as a frame-consistency result and keeps the standard small- β Sagnac observable as the robust experimental limit.

8 Correction-space transfer matrix

The correction hierarchy itself is also a discrete transfer process, but along the correction index rather than along space. Let $q = \sigma\beta$, $r_n = q^n$, and $S_n = \sum_{m=0}^{n-1} q^m$. Then

$$r_{n+1} = qr_n, \quad S_{n+1} = S_n + r_n,$$

so

$$\begin{pmatrix} S_{n+1} \\ r_{n+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 0 & q \end{pmatrix}}_{M_q} \begin{pmatrix} S_n \\ r_n \end{pmatrix}. \quad (20)$$

Starting from $(S_0, r_0)^T = (0, 1)^T$,

$$M_q^N \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} (1 - q^N)/(1 - q) \\ q^N \end{pmatrix}.$$

This transfer matrix is mathematically independent of the physical spatial transfer $U(1)$ product. The two may be organized on the same (j, n) lattice, but they must not be identified.

9 Two distinct convergence mechanisms

For one direction, truncation after order M has remainder

$$R_{\sigma, M} = \frac{(\sigma\beta)^{M+1}}{1 - \sigma\beta}.$$

The one-direction phase error is

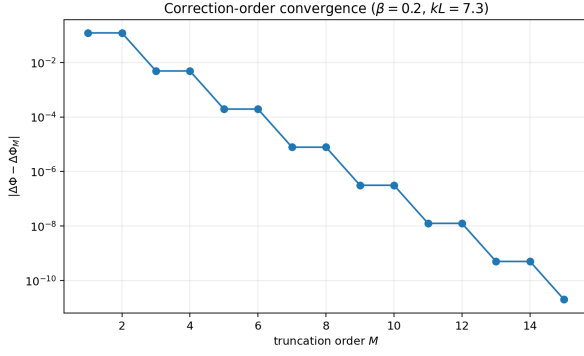
$$|\epsilon_{\sigma, M}| = |k_t L| \left| \frac{(\sigma\beta)^{M+1}}{1 - \sigma\beta} \right|.$$

For the counter-propagating difference and $0 \leq \beta < 1$,

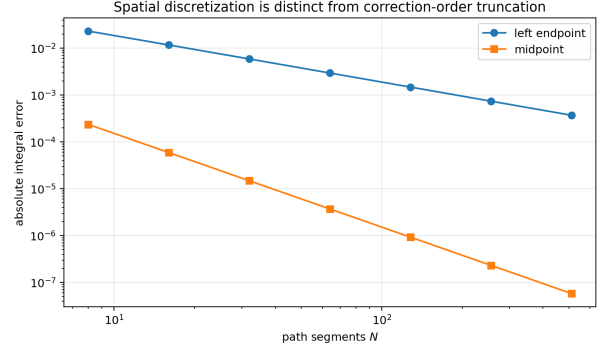
$$|\epsilon_{\Delta, M}| = \begin{cases} \frac{2|k_t L| \beta^{M+1}}{1 - \beta^2}, & M \text{ even}, \\ \frac{2|k_t L| \beta^{M+2}}{1 - \beta^2}, & M \text{ odd}. \end{cases} \quad (21)$$

The repeated plateaus are exact parity consequences.

Spatial quadrature is a different limit. For a smooth nonperiodic test profile, left-endpoint quadrature converges at first order and midpoint quadrature at second order. The familiar limit $(1 + AL/N)^N \rightarrow e^{AL}$ is yet another statement and is not the mechanism used in Theorem 1.



(a) Geometric correction-order convergence.



(b) Spatial quadrature convergence.

Figure 5: The M - and N -limits are different mathematical operations.

10 Time-dependent transfer: the missing phase-rate layer

A detector that identifies events from changes in phase rate requires a time-dependent transfer phase, not merely a constant final phase offset. Let

$$\tau_\sigma(t) = \int_{a(t)}^{b(t)} F_\sigma(s, t) ds, \quad F_\sigma(s, t) = \frac{1}{c[1 - \sigma\beta(s, t)]}. \quad (22)$$

Leibniz' rule gives

$$\dot{\tau}_\sigma(t) = F_\sigma(b(t), t)\dot{b}(t) - F_\sigma(a(t), t)\dot{a}(t) \quad (23)$$

$$+ \int_{a(t)}^{b(t)} \frac{\sigma \partial_t \beta(s, t)}{c[1 - \sigma\beta(s, t)]^2} ds. \quad (24)$$

For fixed endpoints, only the integral term remains.

If the detector/reference angular frequency may vary,

$$\Phi_{\text{transfer}}(t) = \omega_{\text{ref}}(t)\tau_\sigma(t) \quad (25)$$

for the simplest single-frequency model, so

$$\dot{\Phi}_{\text{transfer}} = \dot{\omega}_{\text{ref}} \tau_\sigma + \omega_{\text{ref}} \dot{\tau}_\sigma. \quad (26)$$

A frozen frequency reduces this to $\dot{\Phi} = \omega_{\text{ref}} \dot{\tau}$.

This resolves an ambiguity exposed by the Forward-Lab phase estimator. A single permanent phase rotation can leave the subsequent phase rate unchanged and therefore need not trigger a rate-change detector. A physical M4 prediction must supply the time history of the complex observable or the transfer phase, not only a scalar lag.

11 Reference-frequency discipline: carrier phase is not automatically the JIF phase

For long astrophysical delays, writing $\Phi = 2\pi\nu_{\text{carrier}}\tau$ with a raw radio carrier can be formally true as a local wave relation but operationally useless unless coherence and cycle count are actually preserved across the measurement. The detector-centered phase must therefore declare its reference:

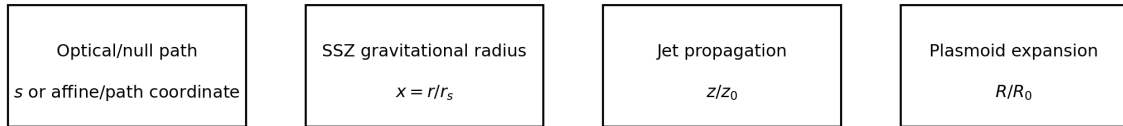
- direct coherent carrier phase, if the instrument genuinely retains it;

- heterodyne/baseband phase relative to a local oscillator;
- cross-spectral phase between measured bands;
- another independently defined JIF phase variable.

The symbol ω_{ref} in this paper means the angular frequency conjugate to the phase record actually used by the detector model. It must not be silently replaced by an astrophysical carrier frequency solely because that carrier appears in a source metadata table.

12 Route and coordinate discipline

The mathematical bridge is exact only after a physical route has been specified. It cannot determine which coordinate in one submodel is identical to a coordinate in another submodel.



These coordinates describe different physical objects and are not interchangeable without an explicit map.

REJECT implicit aliasing

Figure 6: Route-type discipline. These coordinates have different semantics; aliasing them requires an explicit physical map.

12.1 No-coordinate-alias rule

The following identification is *not* licensed by notation alone:

$$\frac{R_{\text{plasmoid}}}{r_s} \stackrel{?}{=} x = \frac{r_{\text{BH-centric}}}{r_s}.$$

In the audited Sgr A* family prior, R/R_0 is the expansion radius of a plasmoid. In the canonical SSZ transport routine, $x = r/r_s$ is the gravitational radial coordinate. They can coincide only if an independent source model explicitly states that the plasmoid size equals its center-of-motion distance from the gravitating center, which the current Forward-Lab does not state. Therefore a numerical calculation obtained by substituting the plasmoid size trajectory directly into the SSZ radial transport integral is classified here as **REJECT**.

12.2 No-double-counting rule

The Forward-Lab observer layer already distinguishes three conceptually different contributions:

1. ballistic moving-source geometry, through $1 \mp \beta \cos i$ applied only to a declared propagation component;
2. an explicit signed path difference, when independently specified;
3. additive observer-transfer terms such as plasma, scattering, lensing, detector, or an explicitly supplied SSZ null correction.

Applying ballistic geometry and an explicit path-difference correction to the same component without a derivation is rejected as double counting.

13 Historical Forward-Lab audit: was the relevant fix already present?

The answer is yes. A source-level audit of the archived public repository changes the interpretation of the earlier exploratory test.

13.1 Ballistic geometry and path routing were already guarded

The module `observer_transfer.py` implements

$$g_{\text{app}} = 1 - \beta \cos i, \quad g_{\text{rec}} = 1 + \beta \cos i,$$

and applies the factor only to `propagation_s`. If both ballistic geometry and `explicit_path_difference_s` are supplied, the code raises a double-counting error. The corresponding test suite checks the approaching/receding limits and explicitly expects the double-count case to fail. The stored path-variant matrix contains separate rows for same-origin/same-path, approaching ballistic, receding ballistic, nearer/farther explicit radio origin, plasma, scattering/detector, signed SSZ-null correction, full identified transfer, and an invalid double-count control.

This means that multiplying the Sgr A* plasmoid expansion time by $(1 - \beta)$ merely because an expansion speed exists is not the historical route semantics. The factor belongs to an identified ballistic displacement along the observer direction, not automatically to the growth of the source radius.

13.2 The dimensional fix was already present

The module `physical_forward_v2.py` contains an explicit correction note: engine, conversion, and opacity terms are already dimensional times in seconds and must not be multiplied again by GM/c^3 . Only the geometric transport term is computed from a distance and speed. The regression test sets a negligible transport distance, an engine delay of 10 s, and $z = 0.1$; the expected observer time is exactly 11 s. Our independent numerical reproduction returns 11 s at both interval endpoints.

13.3 Family-specific coordinates are deliberately different

The v2 family head uses

$$t_{\text{jet}} \propto \frac{z_{\nu} - z_0}{c} \left(\frac{1}{\beta_j} - \cos i \right)$$

for BHXRJ jets, but Sgr A* uses a plasmoid expansion delay

$$t_{\text{pl}} = \frac{R_{\nu} - R_0}{v_{\text{exp}}}.$$

This is direct code evidence that z , R , and the observer ballistic route are intentionally not treated as the same object.

13.4 What the historical M1 path actually was

The later M1 connected ablation uses frozen family-level geometry priors. For Sgr A* the lock contains $x : 1.5 \rightarrow 100$ and local $\beta = 0.1$. Its reported SSZ multiplier is

$$1.0489215194943662.$$

Our independent reintegration of the canonical $D(x)$ profile reproduces this number exactly to double precision. Crucially, the M1 code labels this a *fixed-geometry ablation*; it is not the same thing as claiming that the Sgr A* plasmoid expands physically from $1.5r_s$ to $100r_s$. The previous exploratory interpretation that compared this route with a supposed “actual $x \sim 2 \rightarrow 4$ plasmoid path” was therefore a category error. The historical path was read too literally, and the repo’s own routing semantics already contained the relevant fix.

13.5 Physical M4 exists beyond the synthetic proxy

The historical M1–M3 score correctly excludes M4 because the corresponding ablation uses a synthetic same-signal phase proxy. However, the audited repository also contains a separate physical M4 adapter for typed, frozen complex observables and a real NICER/MAXI J1820+070 cross-spectral artifact. The artifact is target-free with respect to the radio-lag benchmark, uses a locked estimator, and returns `M4_PHYSICAL_NO_ONSET`. This negative result is retained rather than converted into a positive SSZ claim. It is useful technical evidence that the detector-centered phase machinery can ingest a real complex phase observable, but it is not a radio-lag or SSZ validation.

14 Embedding the exact bridge into detector-centered JIF

For each complete contribution/path a , the transfer term may now be represented as

$$\Phi_{\text{transfer}}^{(a)} = \sum_j \sum_n \Delta\Phi_{j,n}^{(a)},$$

with

$$U_{\text{transfer}}^{(a)} = \prod_j \prod_n e^{i\Delta\Phi_{j,n}^{(a)}}.$$

This fills the previously abstract transfer slot of equation (3). It does not replace the source, node, or detector-reference terms. The final observable remains equation (4) at a common detection event.

The resulting architecture is now typed by coupling class:

$$\boxed{\begin{aligned} \Xi, D, g_{\mu\nu} &\rightarrow \begin{cases} \text{vacuum + minimal metric coupling: null HJ/Hamilton/Fermat law,} \\ \text{media/material/extra segment dynamics: separately specified} \end{cases} \\ &\rightarrow \text{routed transfer history} \rightarrow d\Phi_{j,n} \rightarrow T_{j,n} \rightarrow U_{\text{transfer}} \rightarrow \Delta\Phi_D \rightarrow \Delta J_D. \end{aligned}} \quad (27)$$

The arrows from a specified transfer history to phase and $U(1)$ factors are exact. For the present vacuum electromagnetic branch, the project coupling is source-locked rather than refitted here: standard Maxwell dynamics on the declared effective/static SSZ geometry. The remaining model dependence lies in the source/detector boundary-value problem, media and non-vacuum constitutive

physics, rotating/generalized geometry, and any additional SSZ dynamics beyond the source-locked metric-Maxwell branch.

15 Relation to the SSZ radial C^2 bridge

The middle transfer bridge must not be confused with the radial strong-to-weak SSZ bridge. The canonical radial profile joins strong and weak branches across $1.8 \leq x \leq 2.2$ by matching Ξ, Ξ', Ξ'' . The variational bridge paper proves, conditional on fixed branches and endpoint two-jets, that the direct quintic Hermite connection is the unique global minimizer of a squared-third-derivative functional. That theorem chooses a smooth radial field connection. It does not determine a photon transfer route or identify a plasmoid size coordinate with the gravitational radius.

The two bridges may be composed only after the route is declared:

$$\text{radial SSZ profile} \rightarrow D(x), g_{\mu\nu}(x) \rightarrow \text{physical route} \rightarrow \Phi_{\text{transfer}} \rightarrow \Delta J_D.$$

16 Metric-level null information and source-locked Maxwell vacuum completion

The phrase “the SSZ null route remains open” was too broad. The integrated JIF/SSZ manuscript uses the static diagonal line element

$$ds^2 = -c^2 D(r)^2 dT^2 + D(r)^{-2} dr^2 + r^2 d\Omega^2. \quad (28)$$

For a radial null curve, $ds^2 = 0$ and $d\Omega = 0$, hence

$$\boxed{\frac{dr}{dT} = \pm c D^2.} \quad (29)$$

A local static observer uses $d\ell = D^{-1}dr$ and $d\tau = DdT$, so

$$\boxed{\frac{d\ell}{d\tau} = c.} \quad (30)$$

Thus the declared metric already contains definite radial null kinematics and does not imply a locally measured violation of light-speed invariance.

16.1 The vacuum electromagnetic coupling is already source locked

The supplied helper corpus does not leave the vacuum electromagnetic law as an arbitrary new coupling to be chosen in this paper. Its quantitative layer retains Maxwell theory, its curved-spacetime discussion gives the covariant Maxwell system, and its geometric-optics discussion states that rays follow null geodesics of the effective metric while local light speed remains c [18, 20]. The radial-scaling note supplies the complementary gauge picture $d\rho = s(r) dr$ with standard local wave dynamics and explicitly treats this as a geometric reparameterization rather than a material medium [19]. The static SSZ metric used here independently fixes

$$D(r) = \frac{1}{1 + \Xi(r)}, \quad s(r) = \frac{1}{D(r)}.$$

Accordingly, v6 adopts the following *source-locked background-field prescription*:

$$\boxed{\text{fixed SSZ geometry} + \text{standard vacuum Maxwell dynamics} \implies \text{metric geometric optics.}} \quad (31)$$

This is not a derivation of the background geometry from a joint field action and does not include electromagnetic backreaction on Ξ . The broader full-field action/coupling problem therefore remains distinct from the propagation problem solved on the declared background.

16.2 Covariant Maxwell WKB derivation of the eikonal equation

With the source lock in equation (31), the Maxwell-to-ray arrow can be shown explicitly rather than cited as a black box. In vacuum,

$$\nabla_\mu F^{\mu\nu} = 0, \quad \nabla_{[\alpha} F_{\mu\nu]} = 0. \quad (32)$$

Write a high-frequency vector potential as

$$A_\mu = \Re\{[a_\mu + O(\epsilon)] e^{i\Theta/\epsilon}\}, \quad k_\mu \equiv \nabla_\mu \Theta.$$

At leading order, the Maxwell principal symbol is

$$k_\mu (k^\mu a^\nu - k^\nu a^\mu) = 0. \quad (33)$$

The leading Lorenz-gauge/transversality condition is $k_\mu a^\mu = 0$. For a nonzero transverse amplitude, equation (33) therefore requires

$$\boxed{g^{\mu\nu} k_\mu k_\nu = 0.} \quad (34)$$

Because k_μ is a gradient and the Levi-Civita connection is torsion free,

$$k^\nu \nabla_\nu k_\mu = k^\nu \nabla_\mu k_\nu = \frac{1}{2} \nabla_\mu (k^\nu k_\nu) = 0,$$

so k^μ generates affinely parametrized null geodesics. At the next WKB order the amplitude/polarization transport is governed by the usual geometric-optics transport equations; that amplitude layer is not required for the scalar phase theorem developed here. This is the standard metric route from Maxwell wave optics to null-ray optics [15, 16, 17].

For equatorial motion $\theta = \pi/2$, the inverse metric is

$$g^{TT} = -\frac{1}{c^2 D^2}, \quad g^{rr} = D^2, \quad g^{\varphi\varphi} = \frac{1}{r^2}.$$

Stationarity and spherical symmetry allow the separated eikonal

$$\Theta = -\omega T + \ell\varphi + W(r). \quad (35)$$

Defining the impact parameter $b = c\ell/\omega$, equation (34) gives

$$\boxed{W'(r) = \pm \frac{\omega}{cD^2} \sqrt{1 - \frac{D^2 b^2}{r^2}}.} \quad (36)$$

This is an endpoint phase-momentum relation, not yet a detector observable.

16.3 Reconciliation with the radial scaling gauge: s versus s^2

The helper corpus also contains an older, pre-embedding radial-scaling representation with physical radial distance

$$d\ell = d\rho = s(r) dr.$$

That note uses a wave-time symbol before the present metric embedding and at places describes its one-factor path time too loosely as coordinate-level. It cannot be identified simultaneously with the global static coordinate time T of equation (28) and retain $dt = s dr/c$: the metric fixes a two-factor coordinate arrival time. The current metric therefore supplies the authoritative typing. From the metric null law and $s = 1/D$,

$$\frac{dr}{dT} = cD^2 = \frac{cD}{s}, \quad (37)$$

$$\frac{d\ell}{dT} = s \frac{dr}{dT} = cD, \quad (38)$$

$$d\tau_{\text{stat}} = D dT, \quad (39)$$

$$\frac{d\ell}{d\tau_{\text{stat}}} = c. \quad (40)$$

The metric-consistent local static-frame propagation-time element corresponding to the scaled-distance wave picture is therefore

$$\boxed{dt_{\text{sg}} \equiv \frac{d\ell}{c} = \frac{s dr}{c} = D dT.} \quad (41)$$

Thus the two factors that appear in project notes have different types:

$$\boxed{n_\ell \equiv \frac{d\ell}{dr} = s, \quad n_T \equiv c \frac{dT}{dr} = \frac{s}{D} = s^2.} \quad (42)$$

The first is a radial length-scaling/physical-distance factor; the second is the areal-coordinate arrival-time index of the static metric. They agree only in flat space. This typing resolves the apparent s versus s^2 ambiguity without changing locally measured c . It also records a genuine historical notation correction: the older one-factor scaled-distance time must not be substituted directly for the distant metric coordinate T . In v6, the metric-derived T semantics supersede that loose label while preserving the useful spatial-scaling construction.

16.4 Null Hamiltonian and generic equatorial ray equations

The same characteristics follow from

$$\boxed{H(x, p) = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu = 0,} \quad (43)$$

with $p_T = -\omega$, $p_\varphi = \ell$. Explicitly,

$$H = \frac{1}{2} \left[-\frac{p_T^2}{c^2 D^2} + D^2 p_r^2 + \frac{p_\varphi^2}{r^2} \right].$$

Hamilton's equations and the null constraint yield

$$\frac{dT}{d\lambda} = \frac{\omega}{c^2 D^2}, \quad (44)$$

$$\frac{dr}{d\lambda} = \pm \frac{\omega}{c} \sqrt{1 - \frac{D^2 b^2}{r^2}}, \quad (45)$$

$$\frac{d\varphi}{d\lambda} = \frac{\ell}{r^2}, \quad (46)$$

and hence the coordinate ray equations

$$\boxed{\frac{dr}{dT} = \pm cD^2 \sqrt{1 - \frac{D^2 b^2}{r^2}}, \quad \frac{d\varphi}{dr} = \pm \frac{b}{r^2 \sqrt{1 - D^2 b^2 / r^2}}.} \quad (47)$$

For $b = 0$, equation (47) reduces exactly to equation (29). Thus generic vacuum ray kinematics is no longer an unspecified law in the present project lock; it is a consequence of the source-locked Maxwell/effective-metric prescription on the declared static background.

16.5 Optical metric and Fermat arrival-time functional

Setting equation (28) to zero gives

$$c^2 dT^2 = D^{-4} dr^2 + D^{-2} r^2 d\Omega^2.$$

Therefore a spatial route γ has coordinate arrival-time functional

$$\boxed{\Delta T[\gamma] = \frac{1}{c} \int_{\gamma} \sqrt{D^{-4} dr^2 + D^{-2} r^2 d\Omega^2}.} \quad (48)$$

For a static spacetime this is the Riemannian optical/Fermat metric; stationary curves of the arrival time are the spatial projections of null geodesics [16, 17]. Substituting the orbit equation from equation (47) into equation (48) gives

$$\frac{dT}{dr} = \pm \frac{1}{cD^2 \sqrt{1 - D^2 b^2 / r^2}},$$

identical to the Hamiltonian result.

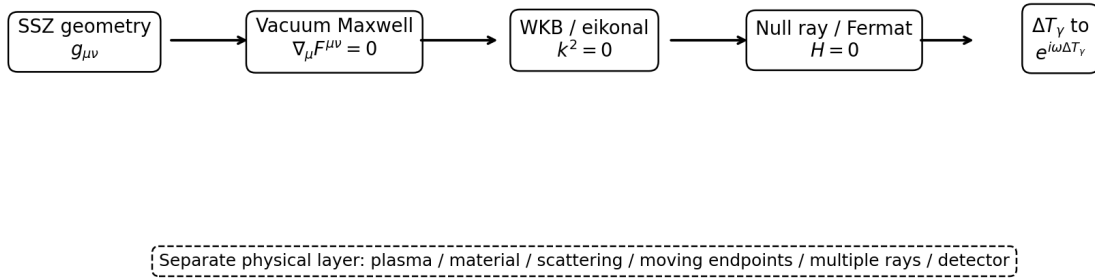


Figure 7: Source-locked vacuum Maxwell/metric closure on the declared background. The metric fixes the geometric-optics null law; media, extra segment dynamics and source/detector boundary-value ownership remain separate layers.

16.6 The null-eikonal safeguard: endpoint phase, not worldline accumulation

A subtle but essential point is that the full spacetime eikonal does *not* accumulate nonzero phase along its own null characteristic. Since $dx^\mu/d\lambda = k^\mu$ up to affine rescaling,

$$\boxed{\frac{d\Theta}{d\lambda} = k_\mu k^\mu = 0.} \quad (49)$$

Consequently, the line integral $\int_\gamma k_\mu dx^\mu$ along the geometric-optics ray is zero when interpreted as the change of the full eikonal. It must not be advertised as a nonzero “photon phase accumulated along the null worldline”.

The measurable transfer relation is instead an endpoint/delay statement. Define the spatial eikonal part

$$W_{\text{sp}} = \ell\varphi + W(r).$$

Along a ray, equation (49) gives

$$\boxed{\Delta W_{\text{sp}} = \omega \Delta T_\gamma.} \quad (50)$$

The same identity follows differentially by substituting equations (36) and (47). If the stationary harmonic convention is $e^{-i\omega T}$ and the detector receives a delayed copy $s_D(T) = s_E(T - \Delta T_\gamma)$, then

$$\boxed{U_\gamma(\omega) = e^{+i\omega \Delta T_\gamma} = e^{i\Delta W_{\text{sp}}}.} \quad (51)$$

Changing the Fourier sign convention changes the sign of the exponent but not the physical relative phase. Equation (51), combined with the detector-centered source/node/reference bookkeeping, is the correct stationary metric-to- $U(1)$ bridge for vacuum propagation.

16.7 Killing frequency versus local detector frequency

The stationary eikonal coefficient ω in equation (35) is conjugate to the Killing time T . A static observer has

$$u_{\text{stat}}^\mu = (D^{-1}, 0, 0, 0)$$

in the present coordinates, so the locally measured angular frequency is

$$\boxed{\omega_{\text{loc}} = -k_\mu u_{\text{stat}}^\mu = \frac{\omega}{D(r)}.} \quad (52)$$

For static emission and detection events,

$$\boxed{\frac{\omega_D}{\omega_E} = \frac{D(r_E)}{D(r_D)}.} \quad (53)$$

This reproduces the detector-centered frequency-transfer rule already used in the integrated JIF layer and makes the clock semantics explicit: ω belongs to T , whereas ω_E and ω_D belong to the corresponding local proper times. The transfer factor in equation (51) therefore does not by itself replace the source and detector oscillator terms in equation (3); those terms are what convert the conserved stationary coordinate frequency into locally referenced measured frequencies.

16.8 Exact radial exterior transfer time

In the exterior branch, with

$$m = GM/c^2, \quad D = (1 + m/r)^{-1},$$

radial outward propagation between fixed radii $r_E < r_D$ obeys

$$\Delta T_{\text{rad}} = \frac{1}{c} \left[(r_D - r_E) + 2m \ln \frac{r_D}{r_E} - m^2 \left(\frac{1}{r_D} - \frac{1}{r_E} \right) \right]. \quad (54)$$

This follows by integrating $dT = dr/(cD^2)$. Under the convention of equation (51), its coordinate-frequency transfer factor is $U_{\text{rad}} = \exp(i\omega \Delta T_{\text{rad}})$. This is a metric-derived conditional result for the declared exterior branch. In the weak-field observable layer it must not be silently substituted for the project’s canonical PPN routing; the final edition keeps this radial transfer calculation separate from the canonical weak-field PPN routing.

For the same exterior branch, the turning condition in equation (47) is $b = r/D = r + m$. Since $d(r/D)/dr = 1$, no finite circular null orbit occurs within that analytic exterior branch itself. Any critical null orbit of the full piecewise SSZ profile must therefore arise from transition/inner structure, consistent with the earlier critical-orbit audit.

16.9 What remains open after the source-locked vacuum derivation

Within the current source lock, the vacuum Maxwell-to-null *differential law* is closed on the declared background by equations (32), (34), (43) and (48); it is no longer an adjustable coupling choice inside this paper. A complete astrophysical prediction still requires a boundary-value problem: source event/worldline, detector worldline, route selection when multiple rays exist, detector reference phase, and ownership of lensing, plasma, dispersion, scattering and instrumental terms. Media require their own constitutive/plasma law and can make the optical problem frequency dependent. A future full field theory may also derive backreaction of electromagnetic stress-energy on Ξ ; that is a different question from propagation on a fixed background. Any additional SSZ segment-transfer dynamics beyond the source-locked Maxwell/metric branch must be introduced as a separately testable extension. The remaining open layer is therefore best written as

$$\boxed{\text{typed boundary/medium problem on the locked vacuum law} \rightarrow \Phi_{\text{transfer}}(t),}$$

not “unknown vacuum light law” or “undecided vacuum coupling”.

17 Canonical weak-field PPN routing and Shapiro normalization

This section is a model-level and method-typing audit; it is not part of the $U(1)$ theorem. The paper keeps only the canonical observable routing used by the project and the still-open requirement that a future unified dynamics derive that completion rather than insert it by hand.

The current canonical documentation is explicit: classify the observable first, then select the method. In the weak-field sector the project lock is

$$\beta_{\text{PPN}} = 1, \quad \gamma_{\text{PPN}} = 1. \quad (55)$$

The routing rule is

$$\text{observable} \rightarrow \text{class} \rightarrow \text{method} = \begin{cases} \text{null/light path} & \rightarrow \text{PPN completion,} \\ \text{static clock/redshift} & \rightarrow \Xi, D, \\ \text{timelike orbit} & \rightarrow \text{PPN}(\beta, \gamma, \dots). \end{cases} \quad (56)$$

The open structural question is therefore precise: derive the canonical weak-field completion in equation (55) from a unified underlying metric/action or from an explicitly defined completion map while preserving the canonical strong-/transition-field architecture. Until that derivation exists, the paper reports the source-locked canonical weak-field layer as an input to the present transfer construction.

For light deflection, the standard first-PN PPN result in terms of the project convention $r_s = 2GM/c^2$ is

$$\alpha = (1 + \gamma) \frac{r_s}{b}, \quad (57)$$

so $\gamma = 1$ gives $\alpha = 2r_s/b = 4GM/(bc^2)$, exactly the factor-of-two completion used by the canonical documentation.

17.1 Shapiro normalization: PPN factor is not the round-trip factor

The audit also found a narrower normalization issue in the helper documentation/code that is independent of whether $\gamma = 1$ is the canonical project value. Standard one-way 1PN Shapiro delay for a static point mass is

$$\Delta t_{1\text{way}} = (1 + \gamma) \frac{GM}{c^3} \ln R = \frac{1 + \gamma}{2} \frac{r_s}{c} \ln R, \quad (58)$$

where R denotes the usual geometric logarithmic ratio (for far conjunction, $R \simeq 4r_1r_2/d^2$). A symmetric radar round trip doubles this:

$$\Delta t_{\text{round}} = 2\Delta t_{1\text{way}} = (1 + \gamma) \frac{r_s}{c} \ln R. \quad (59)$$

Thus, when $r_s = 2GM/c^2$, the expression $(1 + \gamma)(r_s/c) \ln R$ is round-trip-equivalent, not the one-way coefficient. This matters because the project Prime Directive already states that the PPN $(1 + \gamma)$ factor and the independent one-way/round-trip factor must not be conflated. The present paper therefore types them separately and does not use a broad Cassini magnitude check to hide a normalization ambiguity.

For the reference geometry $r_1 = 1 \text{ AU}$, $r_2 = 9.537 \text{ AU}$ and $d = 1.6R_\odot$, the final normalization check gives approximately

$$\Delta t_{1\text{way}} = 132.46 \mu\text{s}, \quad \Delta t_{\text{round}} = 264.92 \mu\text{s},$$

while the Cassini experiment constrains $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ rather than measuring a single geometry-independent absolute delay. This is a reference/typing check, not a new Cassini reanalysis.

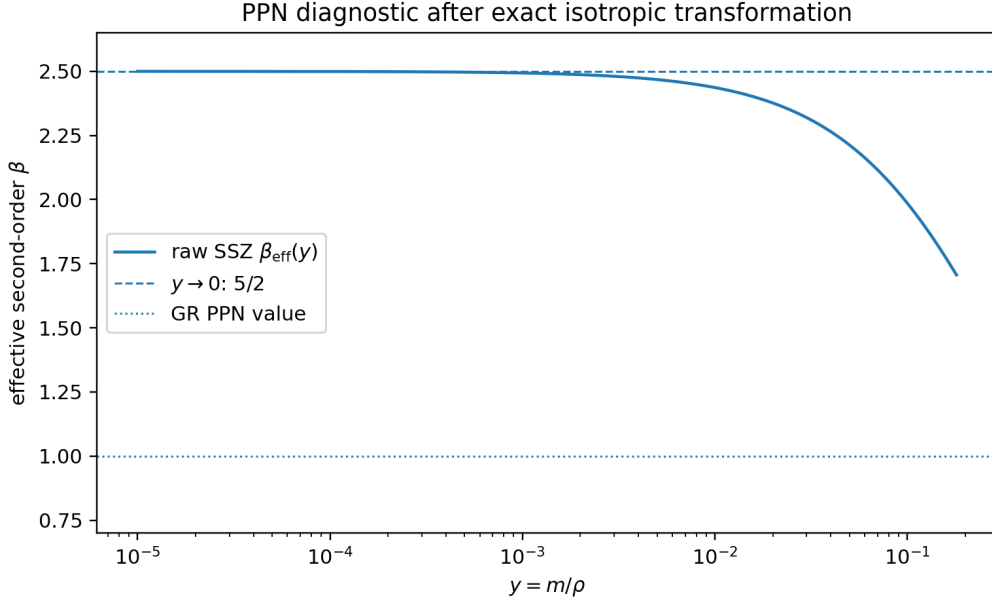


Figure 8: Typed weak-field dispatch and Shapiro normalization. The canonical project-level weak-field observable lock is $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$; Shapiro one-way and round-trip coefficients are separate factors when $r_s = 2GM/c^2$.

18 Relation to finite Fourier-block phase structure: corrected v4 status

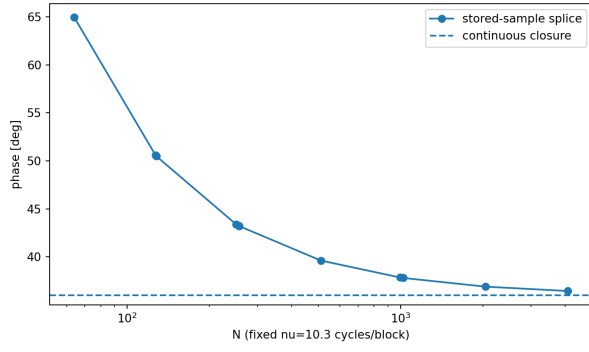
The finite-Fourier-block work is logically independent of the Sagnac/SSZ transfer derivation and is not evidence for it. Its relevance is methodological: different operations can look superficially like the same “phase change” unless their types are fixed. The corrected v4 manuscript proves for a real single tone and any fixed real diagonal quadratic spectral objective

$$J(\phi) = J_0 + 2 \operatorname{Re}(C e^{2i\phi}),$$

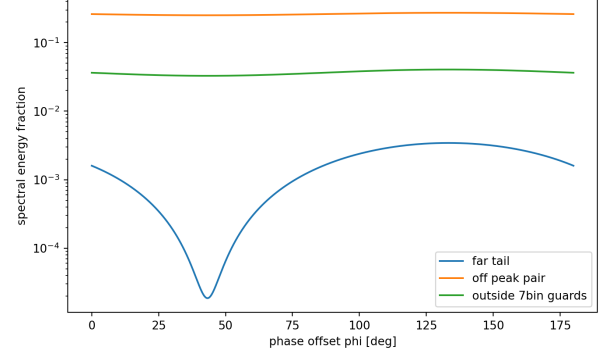
and strictly separates continuous waveform closure, stored-sample splice, and optimization of a declared spectral objective. For the reference case $N = 256$, $\nu = 10.3$ cycles per record,

$$\phi_{\text{cont}} = 36^\circ, \quad \phi_{\text{store}} = 43.2421875^\circ, \quad \phi_{\text{store}} - \phi_{\text{cont}} = \frac{\pi\nu}{N}.$$

The far-tail objective has a max/min phase modulation ratio near 185.12, while two broader diagnostics report only about 1.0878 and 1.2357. The deterministic v4 script passes 19/19 checks and rejects both a universal multi-tone block-start angle and any physically privileged power-of-two grid. The lesson imported here is only one of type discipline: a common phase rotation, a physical time shift, a path coordinate, a sampling boundary, and a detector phase reference are not interchangeable.



(a) Continuous closure versus stored-sample splice.



(b) Objective dependence of the phase sensitivity.

Figure 9: Selected Fourier-v4 cross-checks. They illustrate typed phase operations; they do not validate the Sagnac law or SSZ.

19 Matrix-valued extension and genuine path ordering

The scalar bridge is Abelian. If polarization, Jones vectors, spinors, a rotating local tetrad, or another multicomponent internal space is introduced, the generator may be matrix-valued, $A(s) \in \mathfrak{u}(2), \mathfrak{su}(2), \dots$, and generally $[A(s_1), A(s_2)] \neq 0$. Then the correct object is

$$U = \mathcal{P} \exp \left(\int_{\gamma} A(s) ds \right).$$

This is related in spirit to modern holonomy treatments of Sagnac interferometry, including polarization transport in gravitational-wave backgrounds, but it is a distinct extension and is not claimed as part of the scalar theorem.

20 Validation suite

The validation is deliberately layered. The harmonized lower Middle-Bridge/JIF ledger is locked at 35/35 PASS: 15 core bridge checks, 11 vacuum-null checks, five Maxwell/scaling-gauge checks, and four canonical weak-field/PPN typing checks. The independent wrapper does not arithmetically merge those checks with the new dynamics classifications; instead it verifies the source/regime typing, strong-branch differential fingerprints, radial-direction reversal of the Hermite bridge, Phase-0/Phase-1 reproducibility, and the integrity of the locked 35/35 lower layer. The wrapper returns 15/15 PASS, while the Phase-1 action-search suite is freshly executed and retains its typed PASS/FLAG/REJECTED/CONDITIONAL/OPEN outcomes. The independent Fourier-v4 script remains a logically separate 19/19 phase-typing cross-check, and the historical exterior/variational 23/23 ledger remains an archived cross-audit rather than being misreported as a fresh run.

Before integrating the compactness-response section, the available strong-field audit executable was rerun and reproduced the stored status counts and decisive numerical values. A separate response audit then ran 13/13 PASS checks. In addition to analytic identity checks, it verifies the C^2 change-of-variable matching, integrates χ_u back to the endpoint $\Delta\Xi$, checks all real critical points of the susceptibility polynomial, and cross-checks the global maximum with an independent bounded optimizer and a one-million-point scan. This response suite is mathematical/regression evidence, not an empirical confirmation.

Table 3: Core Middle-Bridge validation ledger. All 15 checks pass in the locked lower layer.

Test	Status	Result
$U(1)$ homomorphism, 2000 random phase sets	PASS	max complex error $< 4 \times 10^{-14}$
Unit-norm preservation, 5000 phases	PASS	$ e^{i\Phi} - 1 $ at floating-point roundoff
Tangent-generator identity $dU/d\Phi = iU$	PASS	max derivative error 1.09×10^{-9} ; below the 10^{-8} validation threshold; radial inner product zero
Continuous generator/product-integral, 120 random profiles	PASS	midpoint product converges to $e^{i \int a ds}$; max error below 3×10^{-7} at fixed $N = 4096$
Propagator composition $U(c, a) = U(c, b)U(b, a)$	PASS	max complex error below 10^{-14}
Sagnac Neumann factorization, 1000 random cases	PASS	exact finite remainder and product identity within floating-point tolerance
Counter-propagating parity/truncation, 1000 cases	PASS	odd-sector identity and pair remainder reproduced
Finite (j, n) reordering, 500 random grids	PASS	max complex discrepancy below 2×10^{-13}
Correction-space transfer matrix	PASS	finite geometric-sum state reproduced over 1000 random cases
Spatial quadrature order	PASS	left rule first order; midpoint second order
Time-dependent fixed-path derivative	PASS	finite difference agrees with analytic derivative
Moving-endpoint Leibniz derivative	PASS	high-precision equality to $< 10^{-45}$ in test units
Coordinate/proper-time phase consistency	PASS	10,000-point finite- β sweep to $\beta = 0.95$; mismatch below 5×10^{-13}
Lifted-phase/wrapped-phase equivalence	PASS	$e^{i(\Phi+2\pi m)} = e^{i\Phi}$ across random integer windings
NICER Crab timing benchmark	PASS	$C(224\text{ s}) = 6632.544791144037$ cycles; left rule first order, midpoint exact for linear $f(t)$ up to roundoff

20.1 Vacuum-null and source-locked coupling validation

The eleven inherited null checks target the metric-to-ray and ray-to-endpoint-phase derivation directly. The lower layer also contains five explicit Maxwell/coupling and scaling-gauge reconciliation checks and four canonical weak-field/PPN typing and normalization checks.

Table 4: Vacuum-null checks. All 11 pass in the locked lower layer.

Test	Status	Result
Inverse-metric algebra	PASS	$g_{\mu\alpha}g^{\alpha\nu} = \delta_{\mu}^{\nu}$ symbolically
Hamilton–Jacobi separation	PASS	separated eikonal satisfies $g^{\mu\nu}\partial_{\mu}\Theta\partial_{\nu}\Theta = 0$ exactly
Generic Hamilton ray equations, 5000 random states	PASS	max $ \delta(dr/dT) = 1.78 \times 10^{-15}$; max orbit-equation error 4.44×10^{-16}
Radial reduction, 1000 random states	PASS	recovers $dr/dT = \pm cD^2$ to 8.88×10^{-16}
Optical/Fermat vs Hamilton arrival-time integrand	PASS	max discrepancy 1.07×10^{-14} over 5000 random states
Null-eikonal constancy	PASS	$k_{\mu}\dot{x}^{\mu} = 0$ to 1.14×10^{-13} in random tests
Endpoint spatial eikonal vs delay	PASS	$dW_{\text{sp}} = \omega dT$ to 2.84×10^{-14}
Raw-exterior radial closed form	PASS	analytic equation (54) vs quadrature, max error 1.42×10^{-14}
Raw-exterior turning relation	PASS	$b = r/D = r + m$ to 7.11×10^{-15} ; $d(r/D)/dr = 1$
Numerical nonradial Hamilton flow	PASS	$ H < 2.24 \times 10^{-15}$; optical-time error 2.84×10^{-14} ; endpoint-eikonal error 5.68×10^{-14}
Stationary delay-transfer convention, 5000 random cases	PASS	$s(T - \tau)/s(T) = e^{+i\omega\tau}$ for $e^{-i\omega T}$, max complex error 4.55×10^{-13}

Table 5: Source-locked Maxwell/scaling-gauge reconciliation checks. All 5 pass in the locked lower layer.

Test	Status	Result
Maxwell WKB principal symbol	PASS	transversality reduces the leading Maxwell symbol to $k^2 a^{\nu} = 0$; nontrivial transverse amplitude selects $k^2 = 0$
Gradient-eikonal geodesic identity	PASS	$k^{\nu}\nabla_{\nu}k_{\mu} = \frac{1}{2}\nabla_{\mu}(k^2) = 0$ for the source-locked Levi-Civita background
Scaling-gauge/metric radial reconciliation, 10,000 random states	PASS	$dt_{\text{sg}} = D dT$, $d\ell/dT = cD$, $d\ell/d\tau_{\text{stat}} = c$ to floating-point precision
Typed optical indices, 10,000 random states	PASS	$n_{\ell} = s$ and $n_T = s/D = s^2$ reproduced independently from the metric null law
Static-observer frequency transfer, 10,000 random endpoint pairs	PASS	$\omega_{\text{loc}} = \omega/D$ and $\omega_D/\omega_E = D_E/D_D$ reproduced exactly to roundoff

Table 6: Canonical weak-field/PPN typing checks. All 4 pass in the locked lower layer.

Test	Status	Result
Canonical PPN parameter lock	PASS	project-typed values $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ are retained as the canonical weak-field observable completion
Lensing normalization	PASS	$(1 + \gamma)r_s/b$ at $\gamma = 1$ equals $4GM/(bc^2)$ to roundoff
Shapiro one-way/round-trip typing	PASS	one-way $(1 + \gamma)r_s/(2c) \ln \mathcal{R}$ and round-trip twice that; the current helper/code coefficient $(1 + \gamma)r_s/c$ is identified as round-trip-equivalent
Cassini-scale normalization	PASS	reference geometry gives $132.46 \mu\text{s}$ one-way and $264.92 \mu\text{s}$ round-trip; $\gamma = 1$ lies within the quoted Cassini constraint

A separate source audit also passes. The supplied electromagnetic helper corpus explicitly retains Maxwell dynamics for quantitative work, states null-geodesic geometric optics for the effective metric, and identifies the radial scaling picture as a geometric reparameterization with invariant locally measured c [18, 19, 20]. The public canonical weak-field documentation explicitly locks $\beta = \gamma = 1$ and the observable-to-method dispatch [21, 22, 23, 24]. The real NICER physical-M4 artifact is frozen before scoring, records `target_radio_lag_used=false` and `fit_to_target_lag=false`, and returns `M4_PHYSICAL_NO_ONSET` [34]. These are provenance/typing audits, not empirical SSZ tests.

20.2 Project-wide validation context: `ssz-all-tests`

The 35/35 ledger above is the executable validation suite for this manuscript; it must not be mistaken for the complete historical SSZ test landscape. The separate repository `error-wtf/ssz-all-tests` documents a final frozen project-wide snapshot of **1296/1296 passed automated checks with zero failures** across twelve listed domains [26, 27, 28, 29]. The integrity file records `total_passed=1296`, `total_failed=0` and `verdict=VERIFIED`. The final status identifies the snapshot as a May-2026 run on Python 3.12/Windows 11.

The master runner is deliberately heterogeneous: it dispatches native repository runners (pytest, dedicated scripts, combined suites and smoke tests) rather than treating all projects as one homogeneous pytest package. Its directly dispatched list contains eleven repositories; the final status additionally contains 103 local `chord-partition` checks, closing the documented total from 1193 to 1296. This detail matters for reproducibility and is one reason the 1296 figure must be read as a regression/validation corpus, not as a count of statistically independent experiments.

Repository-level architecture map. The frozen 1296 snapshot can be mapped onto the present bridge without pretending that every test has the same scientific role. Table 7 is a repository-level classification: the count in a repository is assigned to its dominant architectural role. It is not a test-by-test causal attribution and not an independence claim.

Table 7: Repository-level map of the frozen 1296-check snapshot to the current Middle-Bridge architecture. “Direct” means adjacent to one of the explicit arrows of the present chain; it does not mean an independent proof of that arrow.

Repository/domain	N	Bridge role	Data level
<code>ssz-qubits</code>	184	direct: Ξ, D and $D \rightarrow d\tau \rightarrow d\Phi$	formulas + published benchmarks
<code>ssz-metric-pure</code>	36	direct: $D, s \rightarrow g_{\mu\nu}$, null geometry	mathematics + benchmarks
<code>segmented-calculation-suite</code>	158	direct: Ξ engine, regimes, C^2 bridge, PPN	regression + curated objects
<code>ssz-lensing</code>	279	direct: $g_{\mu\nu} \rightarrow$ null path/WKB-EM phase	synthetic + published lens positions
<code>ssz-trajectories</code>	63	direct: radial/non-radial null-route transport	numerical

Repository/domain	N	Bridge role	Data level
ssz-lagrange	54	direct: Lagrange/Hamilton/effective-potential layer	mathematical
frequency-curvature-validation	82	direct: proper-frequency, Shapiro, clock-loop layer	published experimental values
Unified-Results	147	near: strong-field/data/mass-projection phenomenology	curated astronomy tables
ssz-schumann	178	near: frequency modes and shift diagnostics	mostly unit/synthetic in standard run
g79-cygnus-tests	5	near: nebula/velocity/temperature application	observational CSV/catalogue
segmented-energy chord-partition (local)	7	indirect: N -segment energy context	compiled astronomy parameters
	103	indirect: eigenmodes, φ , geometric identities	mathematical
Total	1296		

At this coarse repository level,

$$856 = 184 + 36 + 158 + 279 + 63 + 54 + 82$$

checks lie immediately adjacent to a bridge layer,

$$330 = 147 + 178 + 5$$

are application/phenomenology-near, and

$$110 = 7 + 103$$

are indirect mathematical/energy-context checks. The partition $856 + 330 + 110 = 1296$ is therefore an *architecture classification*, not a decomposition into independent evidence.

The mapping also clarifies which parts of the present chain predate this paper:

$$\underbrace{\Xi \rightarrow D}_{\text{Calculation Suite, Qubits}}, \quad \underbrace{D, s \rightarrow g_{\mu\nu}}_{\text{Metric Pure}}, \quad \underbrace{g_{\mu\nu} \rightarrow \gamma_{\text{null}}}_{\text{Trajectories, Lagrange, Lensing}}, \quad \underbrace{d\tau, \omega \rightarrow \Phi}_{\text{Qubits, Frequency-Curvature}}.$$

These layers were already separately regression-tested. What the present manuscript newly closes at theorem/architecture level is the explicit additive-phase to $U(1)$ factorization, the scalar continuous generator, the distinction between ray travel time and endpoint transfer phase, and the connection to a detector-relative lifted phase/JIF ledger.

20.3 Data-level evidence hierarchy

Test counts alone do not encode evidential depth. V10 therefore separates three data levels before discussing the newer JIF/NICER layer:

Table 8: Data levels represented in the broader SSZ/JIF validation landscape.

Level	Typical examples	Scientific meaning
A: published benchmark values	Pound–Rebka/Snider, GPS, optical-clock, Cassini/Viking/Mariner-style reference values	real experimental results used at already reduced value/parameter level; not raw instrument streams
B: curated observational tables/features	lens image positions, spectroscopy/emission-line tables, compiled astronomical object parameters	real observations reduced to catalogue/table/feature level; dependence and preprocessing must be reported
C: event/signal-level detector records	NICER event-list to soft/hard light curves to cross-spectrum to complex phase series	closest here to raw-signal analysis; the analysis itself constructs the phase observable and must preserve provenance, gating, unwrapping and estimator locks

The newer `ssz-jif-forward-lab` physical-M4/NICER path belongs to level C and is not part of the frozen 1296 snapshot. In the audited artifact, the source is a public NICER event list, the derived observable is a 1-Hz soft-versus-hard cross-spectral complex signal, the input is frozen before scoring, the radio-lag target is not used, and the locked estimator returns `M4_PHYSICAL_NO_ONSET` [13, 34]. The negative result is retained as a negative result. This is evidence that the detector-centered phase machinery can ingest a real complex signal with explicit provenance; it is not a radio-lag success and not empirical confirmation of SSZ.

For `ssz-schumann`, real-data support exists as a separate capability, but the standard frozen snapshot should not be summarized as “178 real-data tests” when the ordinary path can operate on synthetic/recovery data. This is precisely why the manuscript reports data levels rather than inferring evidence class from repository name or raw test count.

20.4 Validation hierarchy: do not sum unlike suites

The validation picture is therefore intentionally layered:

Table 9: Validation hierarchy reported separately because suites overlap in content, age and abstraction level.

Layer	Status	Interpretation
Legacy cross-repository validation	1296/1296	frozen broad regression/benchmark snapshot across twelve domains; not 1296 independent experiments
Bridge/JIF integrated executable validation	35/35	current paper suite; 31 inherited phase/null/coupling checks plus 4 canonical PPN/normalization typing checks
Exterior/variational cross-audit	archived 23/23	historical independent consistency ledger; overlapping metric identities are rerun in the present suite
Fourier phase validation	fresh 19/19	logically independent phase-typing cross-check; not evidence for SSZ or the Sagnac law
Physical NICER signal audit	<code>M4_PHYSICAL_NO_ONSET</code>	event/signal-level, frozen target-free estimator; technical negative control, not a radio-lag/SSZ score

These suites are reported separately because they overlap in mathematical con-

tent, exercise different abstraction levels, and are not statistically independent. Their counts must not be summed as independent empirical confirmations.

Thus the manuscript does *not* advertise “1332”, “1351”, or any other summed test headline. The broad corpus shows that the new bridge is inserted into a mature regression environment; the paper-specific suite verifies the new logical connections; the NICER audit exercises a qualitatively newer signal-level ingestion path. None of those statements turns test multiplicity into a likelihood ratio for the underlying theory.

20.5 Cross-paper consistency audits

The archived exterior/variational ledger from the v4 cross-audit reports **23/23 PASS**. Its original executable driver is not part of the present mounted analysis bundle, so this edition does not falsely label that historical 23/23 ledger as a fresh rerun. Instead, all exterior identities directly used by this manuscript (radial null law, closed radial integral, raw turning relation, Hamilton flow and canonical PPN typing) are independently covered by the current executable suite. The present paper reports the canonical PPN completion and Shapiro normalization checks used by the manuscript.

The independent Fourier-block validation script *was* rerun unchanged on 25 August 2026 and again returns **19/19 PASS**. These cross-domain ledgers are consistency checks only and are not merged into empirical evidence for the Middle Bridge.

20.6 Correction-order regression

For $\beta = 0.2$ and $kL = 7.3$, the axle-coordinate pair phase in the chosen coordinate-frequency convention is

$$\Delta\Phi_{\text{ax}} = \frac{2kL\beta}{1 - \beta^2} = 3.041666666666665 \text{ rad.}$$

The truncated values are:

M	$\Delta\Phi_M$ [rad]	absolute error [rad]
1	2.9200000000	$1.2166666667 \times 10^{-1}$
2	2.9200000000	$1.2166666667 \times 10^{-1}$
3	3.0368000000	$4.8666666667 \times 10^{-3}$
4	3.0368000000	$4.8666666667 \times 10^{-3}$
5	3.0414720000	$1.9466666667 \times 10^{-4}$
6	3.0414720000	$1.9466666667 \times 10^{-4}$
7	3.0416588800	$7.7866666666 \times 10^{-6}$
8	3.0416588800	$7.7866666666 \times 10^{-6}$

20.7 NICER Crab reference benchmark

The official HEASARC example for Crab ObsID 3013010102 uses

$$f_0 = 29.6095750017 \text{ Hz}, \quad \dot{f} = -368174.57 \times 10^{-15} \text{ Hz s}^{-1},$$

and assigns phase through $C(T) = f_0 T + \frac{1}{2} \dot{f} T^2$. For $T = 224$ s,

$$C_{\text{exact}} = 6632.544791144037 \text{ cycles.}$$

A left-endpoint discretization gives the expected first-order sequence, while midpoint quadrature is exact for the linear frequency model up to floating-point roundoff. For example, the left-rule cycle errors are 9.236764×10^{-6} at $N = 1$, 5.772972×10^{-7} at $N = 16$, and 9.020368×10^{-9} at $N = 1024$.

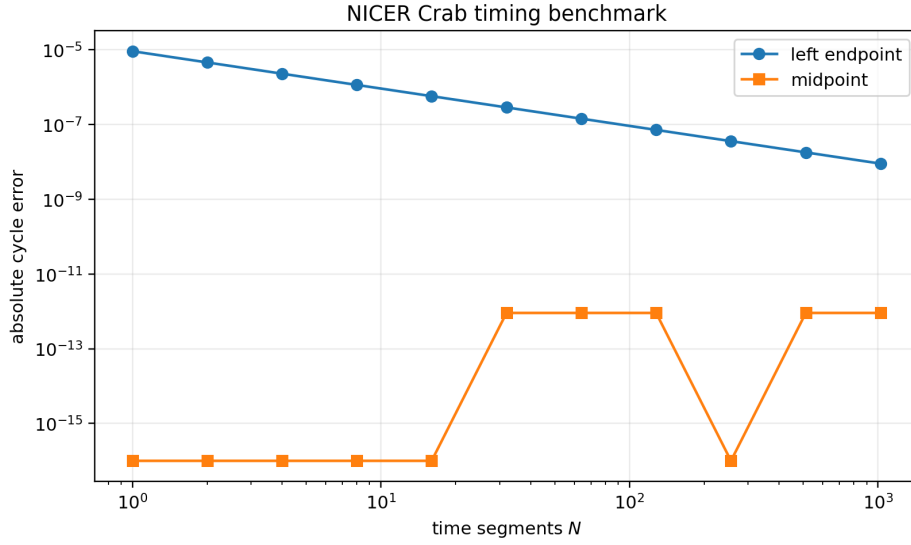


Figure 10: NICER Crab timing benchmark. This validates local phase composition for a known nonconstant phase rate; it does not validate the Sagnac transfer law or SSZ.

20.8 Real physical phase ingestion: MAXI J1820+070

The audited Forward-Lab contains a separate frozen physical-M4 artifact based on public NICER ObsID 1200120106 data released for spectral-timing tutorials. The repository reports 21,817,555 source rows reduced to 57 one-hertz cross-spectral phase samples. The artifact records `pretarget_freeze=true`, `target_radio_lag_used=false`, `fit_to_target_lag=false`, phase reference ownership, the frozen threshold 4.511172510475388, and `PHASE_EVENT_NO_ONSET`; the outer result is therefore `M4_PHYSICAL_NO_ONSET`. The adapter source itself rejects target-contaminated or unfrozen inputs before running the phase estimator [34]. This source-level audit confirms that NICER is not merely a synthetic proxy in the audited repository. It remains a negative technical control, not a radio-lag or SSZ validation.

21 Strong-field response and unified-dynamics audit

21.1 Branch fingerprints and direction-neutral bridge

The strong-first search uses the distinctive information content of the pure branches, not a directional property of the Hermite interpolant. The inner branch

$$\Xi_s(x) = 1 - e^{-\varphi/x} \quad (60)$$

obeys

$$x^2 \Xi'_s + \varphi(1 - \Xi_s) = 0, \quad \varphi = -\frac{x^2 \Xi'_s}{1 - \Xi_s}. \quad (61)$$

The outer branch obeys

$$x \Xi'_o + \Xi_o = 0, \quad 2x \Xi_o = 1. \quad (62)$$

High-precision reconstruction of φ in the final wrapper agrees to better than 10^{-32} on the tested domain. Reversing the normalized bridge coordinate, swapping the endpoint two-jets and reversing the first-derivative orientation reproduces the same quintic profile to approximately 5.6×10^{-16} . Thus the fixed-endpoint variational bridge is mathematically direction-neutral. Strong-first is an epistemic/model-search orientation: retain the more distinctive inner structure as the non-GR anchor and demand that a future dynamics generate the transition and outer/PPN limit.

21.2 Compactness response and the transition-susceptibility theorem

The radial variable itself already contains a dimensionless source-strength coordinate. Define

$$\boxed{C \equiv \frac{r_s}{r} = \frac{1}{x}, \quad u \equiv \frac{GM}{rc^2} = \frac{C}{2} = \frac{1}{2x}.} \quad (63)$$

This is a change of variables, not a new assumption. In these coordinates the two pure branches become

$$\boxed{g_1 : \Xi_o(u) = u, \quad g_2 : \Xi_s(u) = 1 - e^{-2\varphi u}.} \quad (64)$$

The corresponding pure-branch flow laws are

$$\boxed{u \frac{d\Xi_o}{du} = \Xi_o, \quad \frac{d\Xi_s}{du} = 2\varphi(1 - \Xi_s).} \quad (65)$$

Thus the outer branch has linear scale response, while the inner branch has an exponential saturation law. Because the outer g_1 formula is used already for $x > 2.2$ even where the project still labels the physical regime “strong”, the precise statement is *outer linear response versus inner saturating response*; g_1 is not synonymous with the physical weak-field label over its full formula domain. Likewise, $\Xi_s \rightarrow 1$ is only the formal deeper-continuation limit; on the frozen exterior $x \geq 1$ the inner branch reaches $\Xi(1) = 1 - e^{-\varphi} = 0.801711847138$.

The fixed canonical transition

$$1.8 \leq x \leq 2.2 \quad (66)$$

corresponds exactly to

$$\boxed{\frac{5}{22} \leq u \leq \frac{5}{18}} \quad (67)$$

or $5/11 \leq C \leq 5/9$. Its endpoint responses are

$$\Xi\left(\frac{5}{22}\right) = \frac{5}{22} = 0.227272727273, \quad \Xi\left(\frac{5}{18}\right) = 1 - e^{-5\varphi/9} = 0.592986030701. \quad (68)$$

Therefore every differentiable bridge with these same endpoint values and the same compactness window obeys the mean-value theorem: at some u_* ,

$$\boxed{\chi_u(u_*) \equiv \frac{d\Xi}{du} = \frac{1 - e^{-5\varphi/9} - 5/22}{5/18 - 5/22} = 7.241123407889.} \quad (69)$$

This is interpolation-independent. It already exceeds both pure-branch endpoint susceptibilities,

$$\chi_{u,o} = 1, \quad \chi_{u,s} = 2\varphi e^{-5\varphi/9} = 1.317124872442. \quad (70)$$

Hence a transient response amplification is forced by the locked endpoint values and transition width, regardless of which differentiable bridge is chosen.

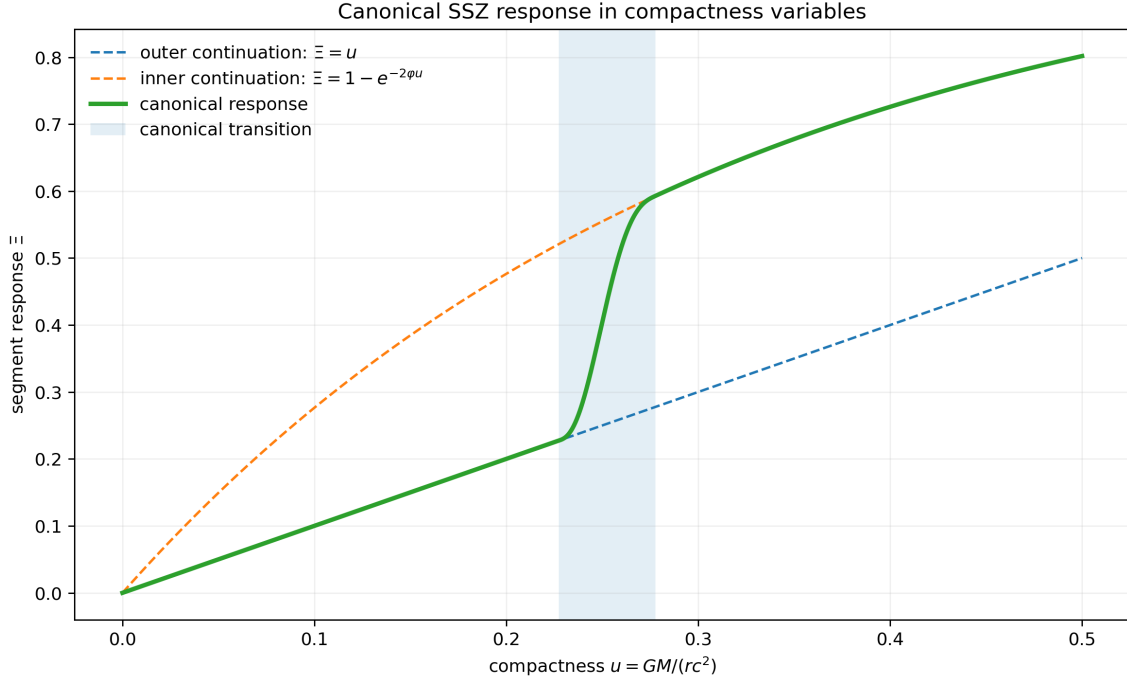


Figure 11: The canonical profile written in the compactness variable $u = GM/(rc^2)$. Dashed curves show the analytic continuations of the two pure branch laws; the solid curve shows the operative piecewise response. The shaded interval is the fixed transition.

For the declared quintic C^2 Hermite bridge, $\chi_u = -2x^2\Xi'(x)$ is a degree-six polynomial in the normalized bridge coordinate after the x^2 factor is included. Checking both endpoints and every real critical point of that stored polynomial gives the global numerical certificate

$$\boxed{\max_{1.8 \leq x \leq 2.2} \chi_u = 12.688510828116} \quad (71)$$

at

$$\boxed{x = 2.008739217432, \quad u = 0.248912350424.} \quad (72)$$

An independent bounded optimizer and a one-million-point scan reproduce the maximum. The susceptibility stays positive throughout the transition, and numerical quadrature of $\chi_u du$ reproduces the endpoint jump in Ξ . The peak is 1.7523 times the interpolation-independent mean-value lower bound and 9.6335 times the larger pure-branch endpoint susceptibility.

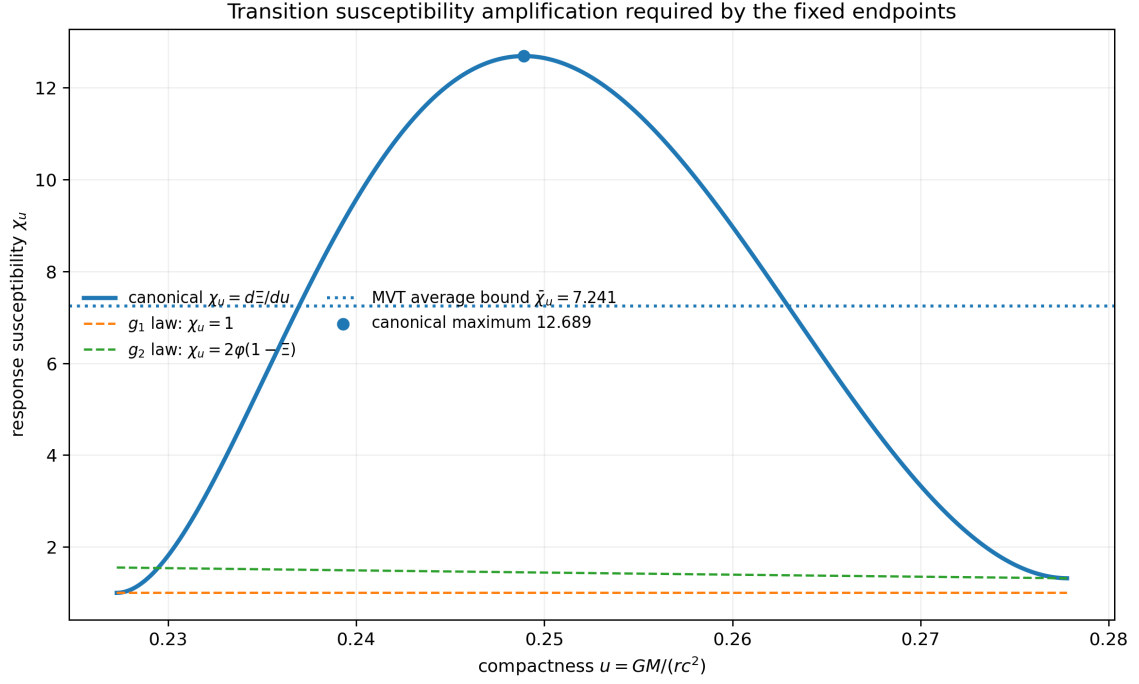


Figure 12: Canonical transition susceptibility $\chi_u = d\Xi/du$. The dotted line is the interpolation-independent mean-value requirement. The actual Hermite bridge overshoots that lower bound and both pure-branch response laws by a large factor before matching the inner and outer endpoint derivatives.

The compactness rewrite therefore does *not* turn the prescribed bridge into either pure-branch flow equation. Defining

$$R_1 = u\Xi_{,u} - \Xi, \quad R_2 = \Xi_{,u} - 2\varphi(1 - \Xi), \quad (73)$$

the outer law has $R_1 = 0$ and the inner law has $R_2 = 0$, but the canonical transition has nonzero residuals in its interior while matching the appropriate endpoint two-jets. This is precisely where a future unified dynamics must do real work rather than merely reuse one pure-branch equation.

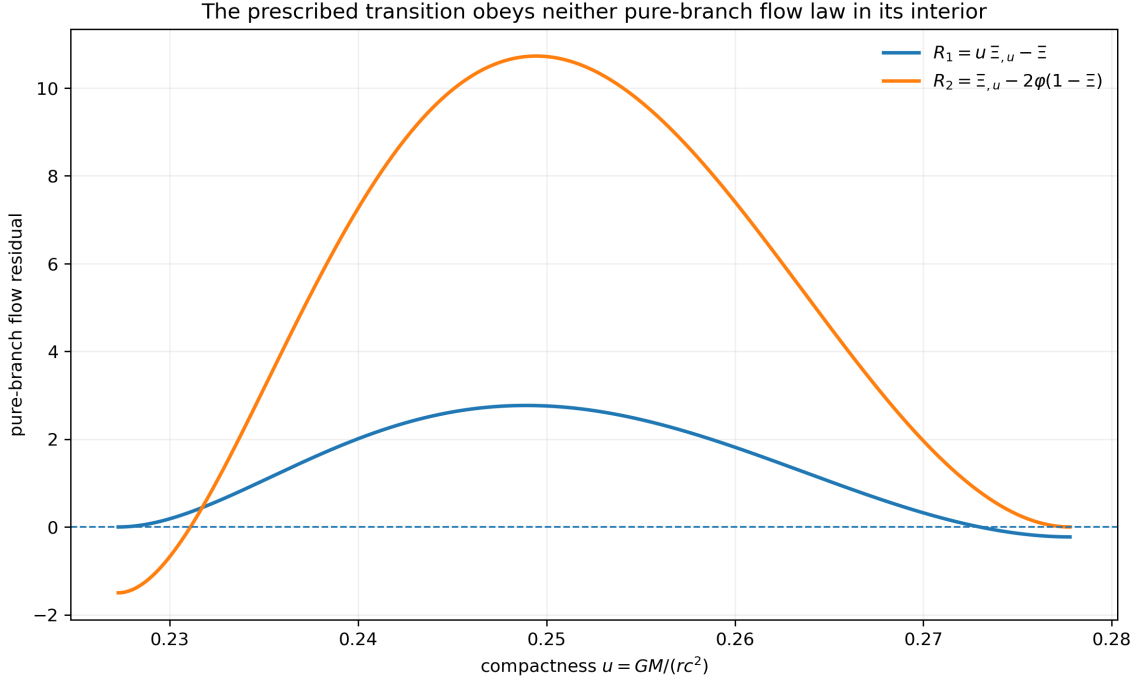


Figure 13: Residuals of the two pure-branch compactness-flow laws evaluated on the prescribed transition. The bridge matches each pure law at the corresponding endpoint but obeys neither law throughout its interior.

The publication-safe interpretation is consequently narrow. The compactness transformation, the two pure-branch flow identities, and the fixed-window susceptibility lower bound are exact. The global maximum quoted for the Hermite bridge is a constructive numerical result with a full critical-point check. The stronger statement that compactness-driven susceptibility amplification is the *physical mechanism* of the regime change remains interpretive until one covariant dynamics derives the branches and transition without point-by-point fitting.

21.3 Einstein-effective source identities

For the declared static gauge $g_{TT} = -A$, $g_{rr} = A^{-1}$, define

$$\hat{\rho} = \frac{1 - A - xA'}{x^2}, \quad \hat{p}_r = -\hat{\rho}, \quad \hat{p}_t = \frac{A''}{2} + \frac{A'}{x}, \quad (74)$$

$$\hat{m}(x) = \frac{x}{2}[1 - A(x)]. \quad (75)$$

Then

$$\boxed{\hat{\rho} = \frac{2\hat{m}'}{x^2}, \quad \hat{p}_t = -\frac{\hat{m}''}{x}, \quad \hat{\rho}' = -\frac{2}{x}(\hat{\rho} + \hat{p}_t).} \quad (76)$$

The equality $\hat{p}_r = -\hat{\rho}$ is a metric-gauge identity of this ansatz, not an independently chosen matter equation of state. Tangential NEC therefore implies $\hat{\rho}' \leq 0$.

21.4 Interpolation-independent endpoint no-go and geometric budget

At the canonical transition endpoints,

$$\hat{m}(1.8) = 0.545334802123639, \quad (77)$$

$$\hat{m}(2.2) = 0.369684499314129, \quad (78)$$

$$\Delta\hat{m} = -0.175650302809510. \quad (79)$$

For every differentiable interpolation sharing those endpoint metric values,

$$\int_{1.8}^{2.2} x^2 \hat{\rho}(x) dx = 2\Delta\hat{m} = -0.351300605619020. \quad (80)$$

The x^2 -weighted mean is

$$\bar{\rho}_w = \frac{6\Delta\hat{m}}{2.2^3 - 1.8^3} = -0.218833433732778, \quad (81)$$

so at least one point satisfies

$$\boxed{\hat{\rho} \leq -0.218833433733.} \quad (82)$$

This is independent of the interpolation variable or polynomial family. The current Ξ -Hermite bridge reaches $\min \hat{\rho} = -0.483042940701$; direct C^2 interpolation of A improves this to -0.467926120437 but cannot remove the endpoint theorem.

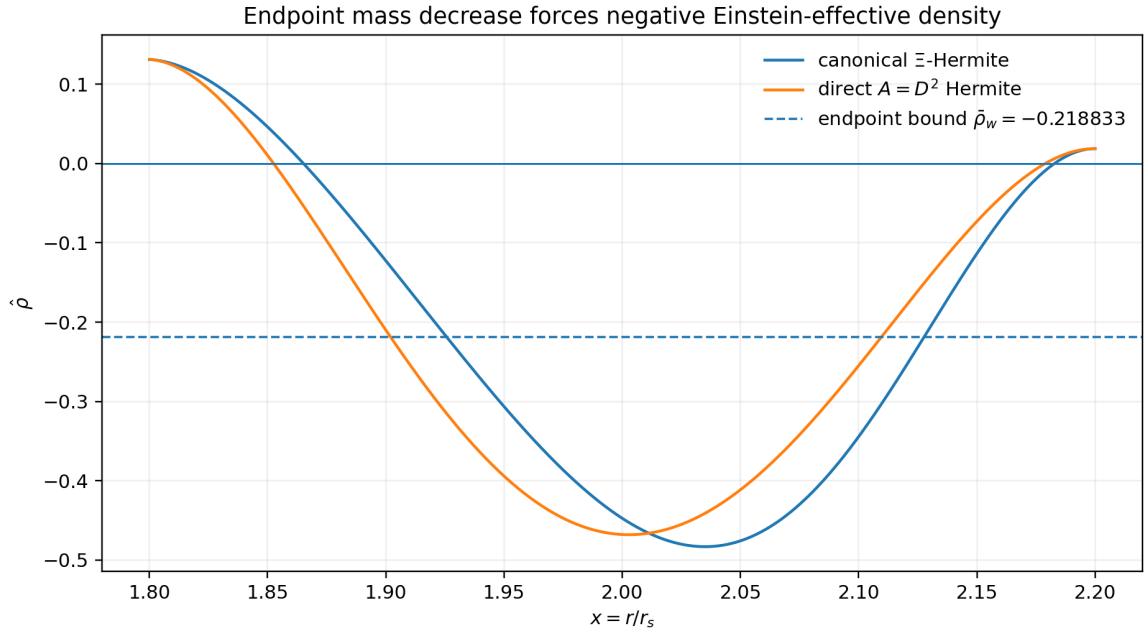


Figure 14: Changing the interpolated variable changes the transition shape but not the endpoint mass no-go. The dashed line is the interpolation-independent weighted-mean bound.

If a future completion retains the same endpoints while requiring nonnegative physical matter density, schematically $\hat{\rho}_{\text{eff}} = \hat{\rho}_{\text{matter}} + \hat{\rho}_{\text{geom}}$ with $\hat{\rho}_{\text{matter}} \geq 0$, then the extra geometric/nonminimal sector must carry at least the negative weighted budget

$$\boxed{\int_{1.8}^{2.2} x^2 \hat{\rho}_{\text{geom}} dx \leq -0.351300605619020.} \quad (83)$$

21.5 Strong-first handoff and the stronger NEC theorem

For outer g_1 ,

$$A_o = \frac{4x^2}{(2x+1)^2}, \quad \hat{m}_o = \frac{x(4x+1)}{2(2x+1)^2}, \quad \lim_{x \rightarrow \infty} \hat{m}_o = \frac{1}{2} \quad (84)$$

from below. The strong branch reaches the same mass normalization at

$$x_* = 1.594811457889, \quad \hat{m}_s(x_*) = \frac{1}{2}. \quad (85)$$

In this metric gauge, $\hat{m} = 1/2$ is algebraically equivalent to $A = 1 - 1/x$, so this is also the known strong-branch $D_{\text{SSZ}} = D_{\text{GR}}$ comparison intersection. If nonnegative Einstein-effective density is required, $\hat{m}$ cannot decrease; hence the current strong branch cannot be retained beyond x_* on the way to the ADM-normalized outer branch. With the outer endpoint fixed at $x_1 = 2.2$, endpoint mass monotonicity gives the stronger finite condition

$$x_0 \leq 1.082828978184. \quad (86)$$

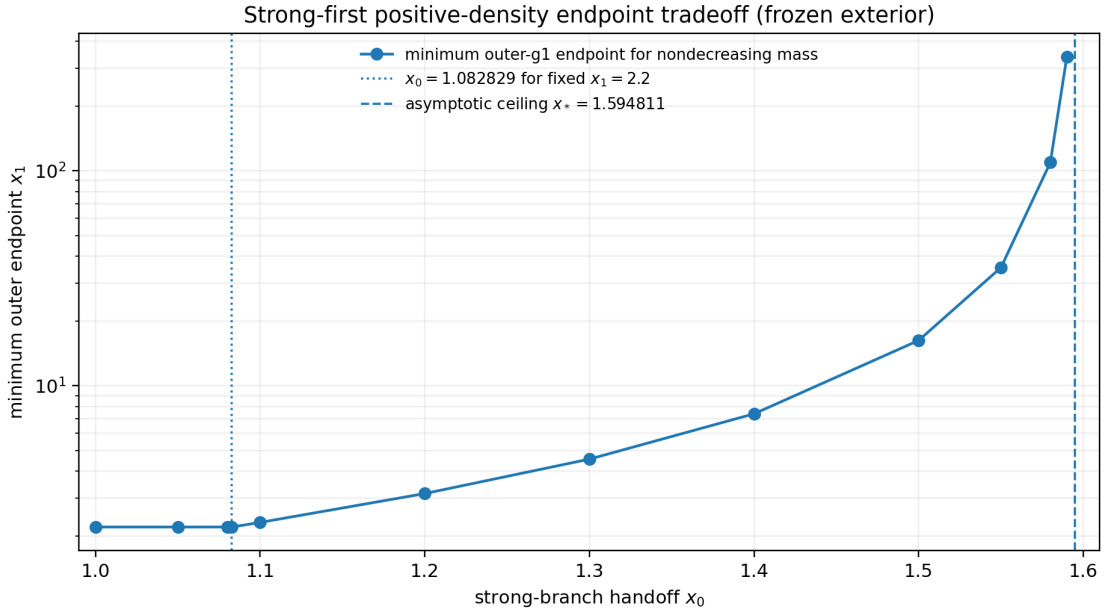


Figure 15: Strong-first endpoint tradeoff for nondecreasing Einstein mass. The required outer endpoint grows rapidly and diverges as $x_0 \rightarrow x_*^-$.

Tangential NEC gives an even sharper result. Since $\hat{\rho} + \hat{p}_t \geq 0$ implies $\hat{\rho}' \leq 0$, matching the fixed outer endpoint at $x_1 = 2.2$ requires $\hat{\rho}(x) \geq \hat{\rho}(2.2)$ throughout the interval. Combining this monotonicity with the required mass integral yields the necessary threshold

$$x_0 \leq 0.977946246457. \quad (87)$$

Because the frozen exterior requires $x_0 \geq 1$,

No everywhere-tangential-NEC ordinary Einstein-matter bridge to $x_1 = 2.2$ exists within the frozen exterior.

This is stronger than the positive-density theorem.

21.6 Constrained transitions: feasibility, not replacement

The endpoint theorem does not say that every SSZ-like interpolation must have negative density. Once the handoff is moved inward, higher-degree mass-space bridges can be constructed under explicit inequality constraints.

Table 10: Constructive feasibility examples. Off-domain examples are continuation diagnostics, not frozen-exterior replacements.

Construction	x_0	x_1	Match	Verified target	Domain
C^2 positive density	1.0	2.2	C^2	$\min \hat{\rho} = 0.00426139$	frozen exterior
C^3 positive density	1.0	2.2	C^3	$\min \hat{\rho} = 0.00133767$	frozen exterior
C^2 density+NEC	0.9	2.2	C^2	$\min(\hat{\rho} + \hat{p}_t) = 0.00552588$	formal off-domain
C^2 DEC+ $p_t \geq 0$	0.5	2.2	C^2	$\min(\hat{\rho} - \hat{p}_t) = 0.00451952$	formal off-domain

The stored floating-point polynomials were checked on a 50,001-point dense grid and at every real critical point of each relevant stored constraint polynomial together with the endpoints. This is a global critical-point numerical certificate for the stored polynomial, stronger than a grid scan but not an exact rational/interval-arithmetic proof.

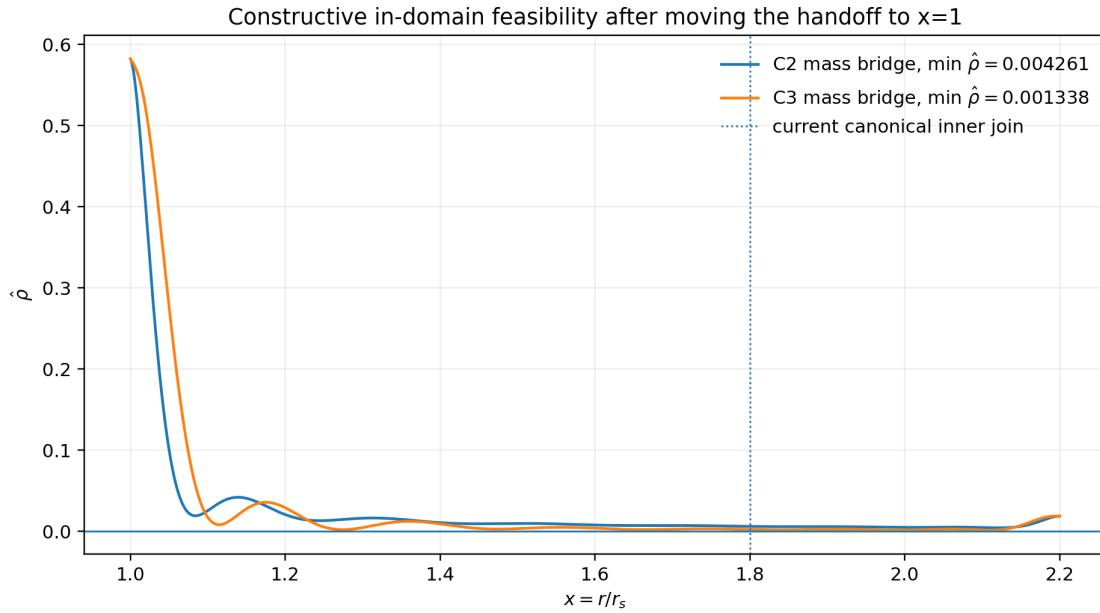


Figure 16: Two in-domain positive-density constructions and off-domain stronger-energy-condition feasibility witnesses. The fixed- x_1 NEC theorem prevents interpreting these as a full ordinary-matter rescue.

21.7 Action-family discrimination

The reconstructed tensor rejects several source classes as sole exact completions:

- any sum of minimally coupled static radial scalar sectors $L_i(X_i, \phi_i)$, which requires $p_t = -\rho$;
- a perfect fluid, which requires $p_r = p_t$;
- pure linear radial Maxwell, which requires $p_t = \rho$ exactly;
- linear Maxwell plus constant Λ , which would require $(\rho - p_t)/2$ constant;

- the simplest uncoupled global-monopole plus fixed-charge linear-Maxwell model, which would require fixed $x^2(\rho - p_t)$ and $x^4 p_t$ scalings that the frozen strong branch does not obey exactly.

These filters do not exclude generalized coupled hedgehog/gauge, scalar-vector-tensor, NED-coupled, nonminimal, or modified-gravity completions.

21.8 Magnetic NED diagnostic with corrected sign convention

For the formal magnetic-NED reconstruction used in the audit, take $L = -\rho$ and $F = x^{-4}$. Then

$$\boxed{2FL_F = -(\rho + p_t), \quad L_F = -\frac{x^4}{2}(\rho + p_t),} \quad (88)$$

and

$$L_F + 2FL_{FF} = L_F - \frac{x}{2} \frac{dL_F}{dx}. \quad (89)$$

For the current transition,

$$L_F = 0 : \quad x = 2.034986003262, \quad 2.199789272082, \quad (90)$$

$$L_F + 2FL_{FF} = 0 : \quad x = 1.927207549701, \quad 2.133996087105. \quad (91)$$

These are in-domain kinetic/characteristic warnings for this one-field reconstruction. On the frozen exterior strong branch $1 \leq x < 1.8$, the corresponding factors keep their Maxwell-connected sign in the tested convention. The separate sign change near $x \simeq 0.43358$ belongs only to formal continuation below $x = 1$ and is not used as an exterior rejection.

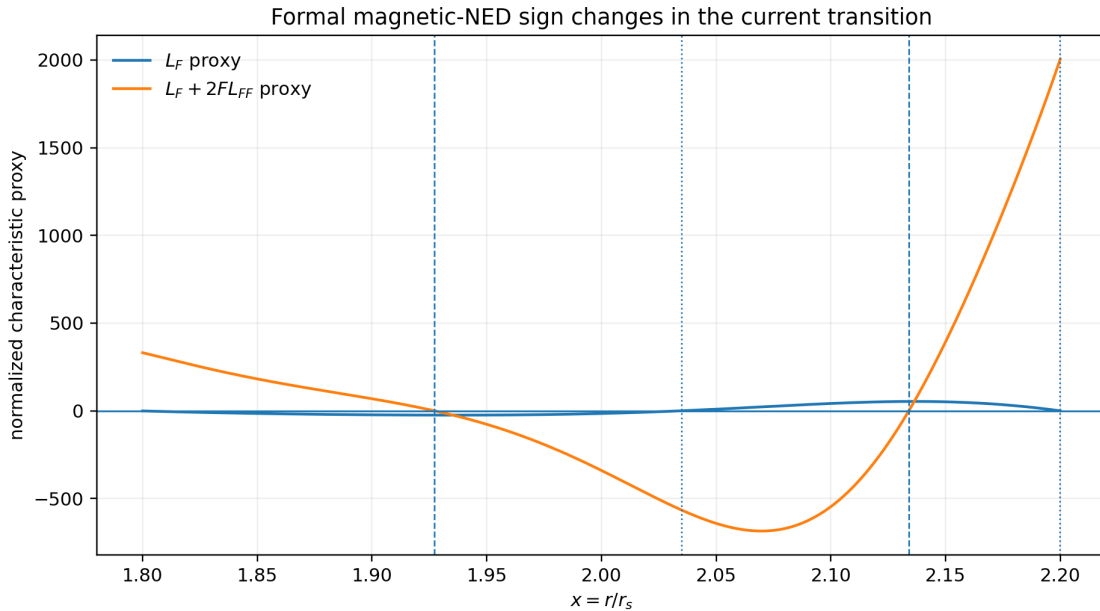


Figure 17: Formal magnetic-NED kinetic/characteristic proxies across the canonical transition, using the corrected sign convention.

21.9 Defect-like to gauge-like structural clue

On each pure frozen branch one may decompose algebraically

$$\rho = \rho_D + \rho_G, \quad \rho_G = p_t, \quad \rho_D = \rho - p_t, \quad (92)$$

with schematic basis stresses $D : \text{diag}(-1, -1, 0, 0)$ and $G : \text{diag}(-1, -1, +1, +1)$. Both coefficients are nonnegative on the pure branches. On outer g_1 ,

$$\hat{\rho}_o = \frac{6x+1}{x^2(2x+1)^3}, \quad \hat{p}_{t,o} = \frac{12}{(2x+1)^4}, \quad x^4 \hat{\rho}_o, x^4 \hat{p}_{t,o} \rightarrow \frac{3}{4}. \quad (93)$$

The outer tail is therefore Maxwell/gauge-like asymptotically. The formal continuation of the inner branch toward $x \rightarrow 0$ is defect/global-monopole-like, but $x < 1$ is outside the frozen exterior and is used only as a model-building clue.

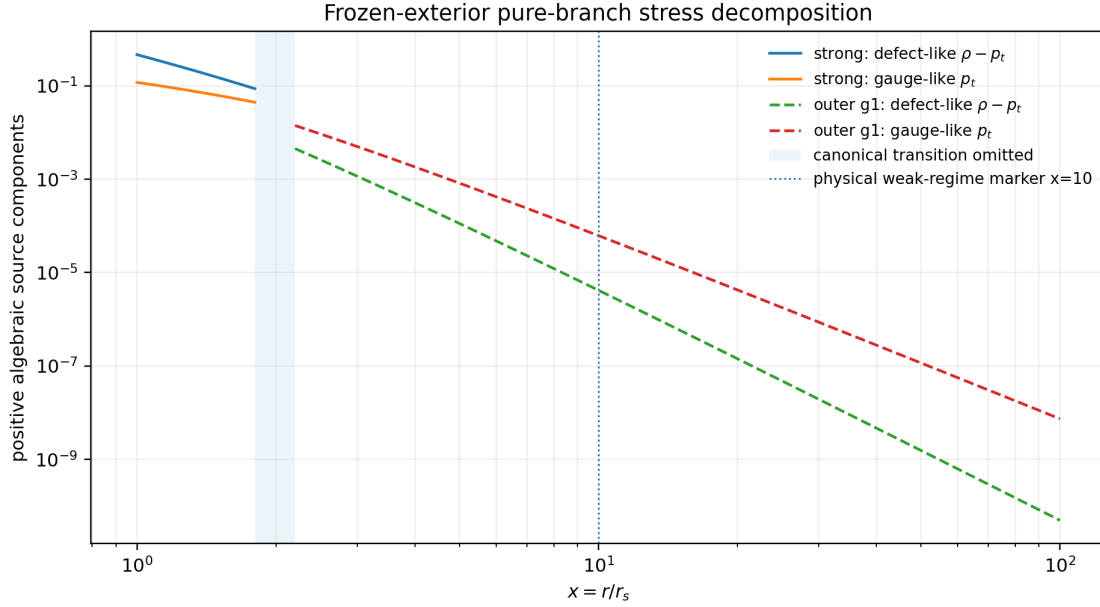


Figure 18: Positive algebraic defect-like and gauge-like source components on the frozen pure branches. The transition is omitted because it leaves this positive cone; $x = 10$ marks a regime label, not a formula switch.

21.10 Regularity filter

The canonical bridge is C^2 but not C^3 . The audit finds

$$|\Delta A^{(3)}| = 140.329289599 \text{ at } x = 1.8, \quad |\Delta A^{(3)}| = 311.400918663 \text{ at } x = 2.2. \quad (94)$$

The Einstein tensor is continuous at the joins, so ordinary second-order metric equations can be treated classically on this background. Generic fourth-order curvature equations are more sensitive and require a smoother bridge, junction sector, or special degeneracy/cancellation. This is a regularity flag, not a blanket rejection of all higher-derivative models. A future global interior must also replace or derive the off-domain continuation because its effective source diverges as $x \rightarrow 0$.

21.11 PPN completion and strong-first interpretation

The publication-safe separation is

$$\boxed{\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1 \text{ as the canonical observable layer,} \quad \text{unified derivation: OPEN.}} \quad (95)$$

The present audit treats this canonical PPN completion as a boundary condition on any future unified dynamics: the same theory that produces the strong branch and its transition must recover

the observable weak-field limit without inserting the PPN parameters by hand.

The pure strong and outer branches have nonnegative Einstein-effective density and lie in the positive defect+gauge cone, whereas the canonical finite transition leaves that cone. This supports

$$\boxed{\text{strong branch} + \text{dynamical transition} \longrightarrow \text{derive outer } g_1/\text{PPN limit.}} \quad (96)$$

The final ledger types this as *INTERPRETIVE_RESULT*, not PASS. The mathematics supports the search orientation; it does not prove that Nature must be “strong-first.”

21.12 Phase-1 validation accounting

The corrected Phase-1 ledger contains 31 typed outcomes:

Status	Count
PASS	18
FLAG	4
REJECTED_CANDIDATE	5
CONDITIONAL_PASS	1
OPEN_CANDIDATE	1
REGULARITY_FLAG	1
INTERPRETIVE_RESULT	1

These are not 31 independent empirical confirmations. Flags and rejections are successful discriminating outcomes; open and interpretive statuses are not passes. The lower Middle-Bridge layer remains separately 35/35 PASS, and the independent final wrapper reports 15/15 consistency checks. None of these heterogeneous layers is arithmetically merged into an empirical success count.

22 Evidence and claim gates

Table 11: Principal statements and their publication-safe status in the present paper.

Statement	Status	Interpretation
Additive phase $\rightarrow U(1)$ factorization	exact	Group-homomorphism identity for absolutely convergent lifted phase.
Scalar generator $U' = iaU$	exact	Unique norm-preserving scalar phase flow under the stated regularity assumptions.
Sagnac Neumann decomposition	exact conditional	Valid for $ \beta < 1$ once the local law is adopted; not a generic strong-gravity law.
(j, n) lattice and two convergences	exact bookkeeping / numerical	Separates route discretization from correction-order truncation.
Finite- β frame split	exact in ideal circular model	Coordinate and detector proper time/frequency must be transformed together.
Maxwell $\rightarrow$ null/Fermat route	exact conditional	Standard vacuum Maxwell WKB on the declared fixed metric; local measured c remains invariant.
Endpoint transfer phase	exact conditional	Detector delay phase; not photon proper-time/worldline accumulation.
Canonical PPN completion	source-locked model layer	$\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$; derivation from a unified action remains open.
Canonical endpoint density no-go	exact within Einstein-effective interpretation	Same endpoint metric values force negative effective density somewhere, independent of interpolation.
Fixed- x_1 tangential-NEC no-go	exact within Einstein-effective interpretation	Everywhere NEC would require $x_0 < 1$, outside the frozen exterior.
Constrained bridges	constructive numerical	Global critical-point certificates for stored floating-point polynomials; not canonical replacements or exact rational proofs.
Minimal scalar/fluid/Maxwell/ Λ /simple monopole+Maxwell sources	rejected as sole exact sources	Algebraic/scaling identities do not match the frozen effective tensor.
One-field magnetic NED	not cleared	Corrected in-domain characteristic-sign crossings remain in the canonical transition.
Compactness rewrite of g_1/g_2	exact reparameterization	$C = 1/x$, $u = C/2$ give $\Xi_o = u$ and $\Xi_s = 1 - e^{-2\varphi u}$ with no new fitted parameter.
Fixed-window susceptibility lower bound	exact for any differentiable fixed-endpoint bridge	Mean-value theorem forces $\chi_u = 7.241123407889$ somewhere between the canonical endpoints.
Canonical Hermite susceptibility maximum	constructive numerical	All real critical points plus endpoint checks give $\max \chi_u = 12.688510828116$; independent optimization and dense scan agree.
Physical response mechanism	interpretive/open	The response language is a dynamical target, not yet a derivation of why the transition occurs.
Strong-first orientation	interpretive result	Supported as an inverse-search strategy; not a physical theorem about Nature's direction.
NICER Crab benchmark	technical/reference	Tests phase composition; not SSZ evidence.
MAXI J1820 M4 no-onset	technical negative control	Real physical phase ingestion; not radio-lag or SSZ validation.
Prospective SSZ confirmation	not claimed	Requires a sealed independent forward prediction and untouched target.

23 Relation to prior work and novelty boundary

The individual mathematical ingredients are standard: the Sagnac effect and its relativistic frame issues have a large literature; geometric/Neumann series are elementary; optical propagation commonly

uses exponential phase or transfer matrices; high-frequency covariant Maxwell theory gives the eikonal/null-geodesic limit; static-spacetime ray paths admit a Fermat/optical-metric formulation; path-ordered exponentials appear for noncommuting transport; and detector-relative phase is standard interferometry [15, 16, 17]. The authors' May 2026 preprint already introduced the geometric correction-series reconstruction and explicitly separated axle coordinate time from detector proper time.

A focused literature search for this revision found extensive Sagnac, geometric-phase, transfer-matrix, and holonomy treatments, including Post's classic review, general-relativistic path treatments, modern geometric-phase reviews, and the 2026 gravitational-holonomy formulation of Javadinezhad and Seraj. We did not identify, in this focused rather than exhaustive search, a prior paper whose central organization is exactly the termwise map

$$(\sigma\beta)^n \rightarrow \Delta t_n \rightarrow \Delta\Phi_n \rightarrow e^{i\Delta\Phi_n}$$

combined with an explicitly independent spatial index j and then inserted into the present detector-centered JIF architecture. Publication-safe novelty wording is therefore limited to: *we provide an explicit termwise correspondence and two-index organization within this project architecture*. No claim is made to discovering exponentials, $U(1)$, transfer matrices, Sagnac physics, or holonomy.

24 Falsifiable next steps

A serious follow-up should preserve the Forward-Lab anti-circular architecture and now move from bridge algebra to dynamics:

1. Declare one concrete second-order or degenerate hedgehog/gauge, scalar-vector-tensor, NED-coupled, or modified-gravity action family before fitting the transition.
2. Derive its Euler–Lagrange equations and test whether the pure compactness-flow targets $u\Xi_{,u} = \Xi$ on the outer branch and $\Xi_{,u} = 2\varphi(1 - \Xi)$ on the inner branch emerge without point-by-point fitting.
3. Generate the transition dynamically and test whether the derived solution satisfies the exact fixed-endpoint susceptibility budget. If the canonical endpoints and width are retained, the solution must attain $\chi_u \geq 7.241123407889$ somewhere; if it does not, the endpoint architecture or transition width must change.
4. Derive the canonical weak-field observable completion $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$ from the same action or a rigorously defined completion map.
5. Derive the quadratic perturbation action and test ghosts, gradient signs, hyperbolicity, characteristic cones, and whether the transition amplification introduces a pathological kinetic/gradient scale.
6. If a global interior is claimed, replace or derive the off-domain $x < 1$ continuation and establish regularity.
7. Produce a distinct strong-field observable; freeze source, transfer, detector, estimator and uncertainty model; seal the prediction; only then open independent target data.

The intended pipeline is

declare action $\rightarrow$ derive solution/transition $\rightarrow$ derive PPN $\rightarrow$ test stability $\rightarrow$ seal prediction $\rightarrow$ open target	(97)
--	------

25 Discussion

The present paper combines the exact Middle-Bridge/JIF/Maxwell route with the corrected endpoint theorems, source-family filters, domain discipline, and the compactness-response audit. The purpose of the integration is not to inflate the evidence count, but to keep the phase-transfer mathematics, the radial transition constraints, and the open unified-dynamics problem in one claim-typed framework.

The central mathematical bridge is modest but clean: additive phase is represented multiplicatively in $U(1)$, and a real scalar generator integrates to a unitary phase flow. The geometric/Neumann series is one conditional source of additive transfer corrections, not the origin of the exponential map. This distinction prevents a category error that had obscured the middle arrow. The two-index (j, n) lattice and the separate convergence limits further prevent numerical segmentation from being confused with correction order.

The propagation layer is now equally typed. On the declared static vacuum background, Maxwell WKB fixes null characteristics, the optical metric fixes the Fermat route, and stationary delay produces an endpoint phase. The same construction shows why a nonzero detector phase delay does not imply nonzero phase accumulation along a null worldline. Coordinate speed, local measured speed, Killing frequency and detector-local frequency are different objects with explicit transformation rules. Weak-field PPN routing remains separately source-locked until a unified dynamics derives the same observable completion.

The strongest conceptual advance of the dynamics audit is that the canonical transition problem is not an interpolation-choice problem. The endpoint mass decrease fixes a negative weighted Einstein-effective density budget before any polynomial is chosen. The strong branch already overshoots the outer ADM-normalized mass at $x = 1.8$, and the stronger NEC theorem shows that moving the handoff merely to the frozen boundary $x = 1$ still cannot produce an everywhere-tangential-NEC Einstein-matter bridge to the fixed outer endpoint $x_1 = 2.2$. Consequently, if the current inner branch and current outer endpoint architecture are retained, a future completion must either supply an extra geometric/nonminimal sector or dynamically alter the transition/endpoints.

The compactness-response rewrite adds an independent geometric constraint of a different type. It does not depend on the Einstein-effective matter interpretation. Once $u = GM/(rc^2)$ is used, the two pure branches become a linear outer law and an exponential inner law, while the fixed endpoints force a large intermediate susceptibility by the mean-value theorem. The canonical Hermite bridge amplifies that susceptibility even further. This makes the missing dynamics more concrete: a successful theory must explain not merely why there are two branch formulas, but how a narrow transition carries the required response change without introducing an unacceptable instability. Conversely, a future dynamics is free to avoid the numerical amplification only by deriving different transition endpoints or a different transition width.

The constrained bridge constructions remain useful precisely because they show what the no-go does not prove. Positive density is achievable after moving the handoff, so the obstruction is not “Hermite” or “interpolation” in the abstract. But the stronger energy-condition gate prevents turning those examples into a claim that ordinary matter has solved the transition. Likewise, the defect-to-gauge decomposition and NED diagnostics are action-search filters, not matter identifications. They reduce the candidate space while leaving the fundamental inverse problem open.

Validation language is deliberately conservative. Exact mathematics, software regression, technical reference benchmarks, physical negative controls and empirical forward evidence are separate categories. The 35/35 lower-layer suite, 15/15 wrapper, 31 typed Phase-1 outcomes, 19/19 Fourier checks and historical 1296/1296 project snapshot are not summed. A future positive claim must come from a prediction frozen before the final target observable is inspected.

26 Conclusion

This paper closes the Middle Transfer Bridge under its stated assumptions and makes the next unresolved problem more precise. The scalar transfer core is exact: $\sum \Delta\Phi_n$ and $\prod e^{i\Delta\Phi_n}$ are the additive and multiplicative representations of the same lifted phase, with $U' = iaU$ providing the continuous generator. The Sagnac Neumann hierarchy is conditional on the declared local law and $|\beta| < 1$; strong gravitational field by itself is not the convergence parameter. Spatial path segmentation remains distinct from correction order, and the ideal finite- β frame split requires conjugate clock/frequency semantics.

For vacuum propagation on the declared static geometry, standard Maxwell WKB, the null Hamiltonian and Fermat principle give one consistent null-route layer while preserving locally measured c . The observable stationary transfer is an endpoint delay phase, not photon proper time. This layer can feed the detector-centered JIF construction once the source, route, node and detector reference are fully typed.

The strong-field dynamics result is sharper. The canonical $1.8 \rightarrow 2.2$ endpoints force a negative Einstein-effective weighted mass budget independently of interpolation. The strong branch reaches the outer ADM mass normalization at $x_* = 1.594811457889$; with $x_1 = 2.2$, positive density requires $x_0 \leq 1.082828978184$, and everywhere tangential NEC would require $x_0 \leq 0.977946246457$, outside the frozen exterior. Thus the present transition cannot be sourced by ordinary everywhere-NEC Einstein matter without changing the endpoint architecture or adding a nonminimal/geometric sector.

The compactness-response audit supplies a second exact transition constraint. With $u = GM/(rc^2)$, the outer branch is $\Xi = u$ and the inner branch is $\Xi = 1 - e^{-2\varphi u}$. The fixed canonical endpoints force an average transition susceptibility $\bar{\chi}_u = 7.241123407889$, while the actual Hermite bridge reaches 12.688510828116. The lower bound is interpolation-independent; the Hermite maximum is constructively and independently verified. These results do not derive the physical mechanism, but they give any future dynamics a precise response target.

The action search is narrowed but not solved. Simple scalar, fluid, linear-Maxwell, constant- Λ , and standard uncoupled monopole+Maxwell sources fail exact gates. One-field magnetic NED remains problematic in the current transition. The positive defect-like/gauge-like branch decomposition motivates generalized hedgehog/gauge and second-order/degenerate scalar-vector-tensor families, but only an explicit action can decide the issue. That action must derive the strong branch, generate the transition, recover $\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$, pass quadratic stability, and produce a sealed prospective strong-field discriminator.

The publication-safe endpoint is: the phase bridge is closed, the transition constraints are substantially sharpened, the candidate dynamics are restricted, and the remaining claims are left open exactly where the present evidence leaves them open.

A Proof of uniform correction convergence

Assume $|\beta(s)| \leq \beta_{\max} < 1$ on a path of finite length. Then

$$|k_t(\sigma\beta(s))^n| \leq |k_t|\beta_{\max}^n.$$

The majorant series $\sum_n |k_t|\beta_{\max}^n = |k_t|/(1 - \beta_{\max})$ converges. By the Weierstrass M-test, the correction-order series converges uniformly. Thus, under the finite-path regularity assumptions,

$$\int_{\gamma} \sum_n k_t(\sigma\beta(s))^n ds = \sum_n \int_{\gamma} k_t(\sigma\beta(s))^n ds.$$

B Derivation of the pair truncation error

From equation (15),

$$\Delta\Phi = 2k_t L \sum_{q=0}^{\infty} \beta^{2q+1}.$$

If M is odd, the next surviving power is $M + 2$, yielding

$$\epsilon_{\Delta,M} = \frac{2k_t L \beta^{M+2}}{1 - \beta^2}.$$

If M is even, the next surviving power is $M + 1$, yielding

$$\epsilon_{\Delta,M} = \frac{2k_t L \beta^{M+1}}{1 - \beta^2}.$$

C Moving-endpoint derivative check

For $\tau(t) = \int_{a(t)}^{b(t)} F(s, t) ds$, the derivative is

$$\dot{\tau} = F(b, t)\dot{b} - F(a, t)\dot{a} + \int_a^b \partial_t F(s, t) ds.$$

With $F_\sigma = [c(1 - \sigma\beta)]^{-1}$,

$$\partial_t F_\sigma = \frac{\sigma \partial_t \beta}{c(1 - \sigma\beta)^2}.$$

The numerical regression used a smooth polynomial-affine test field and moving endpoints; central finite differences agreed with the analytic derivative to a maximum absolute discrepancy 5.2711×10^{-82} in the test units.

D Historical software-source audit

Source	Audited role
<code>src/observer_transfer.py</code>	Implements $1 \mp \beta \cos i$ only on propagation, separate signed path difference and transfer terms, and double-count rejection.
<code>tests/test_observer_transfer_v1_5.py</code>	Regression checks for approaching/receding factors, signed path difference, redshift routing, and double-count failure.
<code>results/path_variant_matrix_v1_5.csv</code>	Stores independent route variants and an explicitly invalid double-count control.
<code>src/physical_forward_v2.py</code>	Contains the dimensional fix: dimensional engine/conversion/opacity seconds are not multiplied by GM/c^3 .
<code>tests/test_physical_forward_v2.py</code>	Locks the dimensional regression to 11 s for a 10 s source term at $z = 0.1$ when transport is negligible.
<code>src/ssz_jif_forward_v2.py</code>	Separates BHXRB jet propagation, Sgr A* plasmoid expansion, and TDE delay heads.
<code>src/ssz_native_closure_v3.py</code>	Defines canonical $x = r/r_s$ SSZ transport and explicitly distinguishes local material β from SSZ model quantities.
<code>src/v2_ssz_jif_ablation.py</code>	Uses fixed family geometry priors as connected ablations; does not claim those priors are source trajectories.
<code>src/jif/physical_m4.py</code>	Physical typed complex-observable adapter with frozen estimator/calibration gates.
<code>results/physical_m4_real/MAXIJ1820_NICER_CROSS_SPECTRAL_1HZ.json</code>	Real public complex phase ingestion; retained no-onset result; no radio-lag benchmark score.

E Minimal reproducibility pseudocode

```
# correction-order phase on one declared path segment
for j in segments:
    for n in orders:
        dphi[j,n] = k_ref * ds[j] * (sigma * beta[j])**n
        T[j,n] = exp(1j * dphi[j,n])

Phi_transfer = sum(dphi[j,n] for j in segments for n in orders)
U_transfer    = product(T[j,n] for j in segments for n in orders)
assert close(U_transfer, exp(1j * Phi_transfer))

# detector-centered composition
Phi_D_a = Phi_emit_a + Phi_transfer_a + sum(Phi_nodes_a) + Phi_detector_a
Phi_D_b = Phi_emit_b + Phi_transfer_b + sum(Phi_nodes_b) + Phi_detector_b
Delta_J = unwrap(Phi_D_b - Phi_D_a) / (2*pi)

# guardrails
assert route_coordinate_is_typed
assert not alias(R_over_R0, r_over_rs) unless explicit_physical_map
assert not (ballistic_geometry and explicit_path_difference_for_same_component)
assert frequency_reference_is_conjugate_to_phase_time_variable
```

F Generator-flow proof and geometric interpretation

For $U' = ia(s)U$ with real a , multiplication by ia is anti-Hermitian in the one-dimensional complex Hilbert space. Hence

$$\frac{d}{ds}(U^*U) = U'^*U + U^*U' = (-iaU^*)U + U^*(iaU) = 0.$$

The flow therefore remains on the unit circle. Writing $U = e^{i\Phi}$ gives $\Phi' = a$ wherever the lift is chosen continuously. Conversely, integrating the phase first gives $\Phi(s) - \Phi(s_0) = \int a ds$ and exponentiating reproduces the same solution. This establishes the equivalence of local phase density, lifted phase accumulation, and scalar unitary transport.

G Source-locked Maxwell/null-law algebra and phase safeguard

For the equatorial metric $g_{\mu\nu} = \text{diag}(-c^2D^2, D^{-2}, r^2)$, the inverse is $\text{diag}[-(c^2D^2)^{-1}, D^2, r^{-2}]$. Substituting $\Theta = -\omega T + \ell\varphi + W(r)$ into $g^{\mu\nu}\partial_\mu\Theta\partial_\nu\Theta = 0$ gives equation (36). Hamilton's equations from equation (43) then give equation (47).

For the endpoint relation, write $W_{\text{sp}} = \ell\varphi + W$. Along a ray,

$$\frac{dW_{\text{sp}}}{dr} = W'(r) + \ell\frac{d\varphi}{dr} = \frac{\omega}{cD^2\sqrt{1-D^2b^2/r^2}} = \omega\frac{dT}{dr}.$$

Thus $\Delta W_{\text{sp}} = \omega\Delta T_\gamma$, while the full eikonal obeys $\Delta\Theta = -\omega\Delta T_\gamma + \Delta W_{\text{sp}} = 0$. This explicitly distinguishes a detector transfer phase from a fictitious accumulated photon proper phase.

For the exterior $D = r/(r+m)$, radial integration yields equation (54) because $D^{-2} = (1+m/r)^2$. The same branch has $r/D = r+m$, so the null turning function is strictly monotone. These formulas

were checked symbolically and numerically in the current executable ledger.

H Strong-first reproducibility formulas

For any $A(x)$ in the present metric gauge,

$$\hat{\rho} = \frac{1 - A - xA'}{x^2}, \quad \hat{p}_r = -\hat{\rho}, \quad \hat{p}_t = \frac{A''}{2} + \frac{A'}{x}, \quad (98)$$

$$\hat{m} = \frac{x}{2}(1 - A), \quad \hat{\rho} = \frac{2\hat{m}'}{x^2}, \quad \hat{p}_t = -\frac{\hat{m}''}{x}, \quad (99)$$

$$\hat{\rho}' = -\frac{2}{x}(\hat{\rho} + \hat{p}_t). \quad (100)$$

For outer g_1 , $\Xi_o = 1/(2x)$,

$$A_o = \frac{4x^2}{(2x + 1)^2}, \quad (101)$$

$$\hat{m}_o = \frac{x(4x + 1)}{2(2x + 1)^2}, \quad (102)$$

$$\hat{\rho}_o = \frac{6x + 1}{x^2(2x + 1)^3}, \quad (103)$$

$$\hat{p}_{t,o} = \frac{12}{(2x + 1)^4}. \quad (104)$$

The endpoint identity is

$$\int_{x_0}^{x_1} x^2 \hat{\rho} dx = 2[\hat{m}(x_1) - \hat{m}(x_0)]. \quad (105)$$

For the magnetic-NED diagnostic with $L = -\rho$ and $F = x^{-4}$,

$$2FL_F = -(\rho + p_t), \quad L_F + 2FL_{FF} = L_F - \frac{x}{2}L'_F. \quad (106)$$

I Compactness-response reproducibility formulas

For the response audit, the transition variable is $t = (x - 1.8)/0.4$ and the canonical bridge is the quintic Hermite polynomial that matches Ξ , Ξ' , and Ξ'' to the two pure branches. Under $C = 1/x$ and $u = C/2$,

$$\frac{d\Xi}{dC} = -x^2\Xi'(x), \quad \frac{d\Xi}{du} = -2x^2\Xi'(x), \quad (107)$$

and

$$\frac{d^2\Xi}{dC^2} = 2x^3\Xi'(x) + x^4\Xi''(x). \quad (108)$$

Therefore C^2 matching in x remains C^2 matching in C and u because $x \mapsto 1/x$ is smooth and invertible on the frozen exterior. The global maximum of χ_u is certified numerically by solving the polynomial critical-point equation for all real roots in $t \in [0, 1]$ and evaluating those roots together with both endpoints. A separate bounded scalar optimization and a one-million-point scan agree with the critical-point result, and trapezoidal integration on 200,001 compactness samples gives

$$\int_{5/22}^{5/18} \chi_u du = \Xi(5/18) - \Xi(5/22) \quad (109)$$

to better than 2×10^{-10} in the audit run.

J AI assistance disclosure

This manuscript was prepared with AI assistance under human-directed source locks and explicit computational checks. AI was used for algebraic review, source comparison, code auditing, symbolic verification, source reconciliation, figure recovery, document integration and typesetting. The paper does not treat AI-generated wording or test counts as evidence. Claims are restricted to the declared assumptions, source-locked profile, mathematical derivations, and validation artifacts described in the paper; unresolved dynamics, stability, interior and empirical gates remain explicitly open.

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